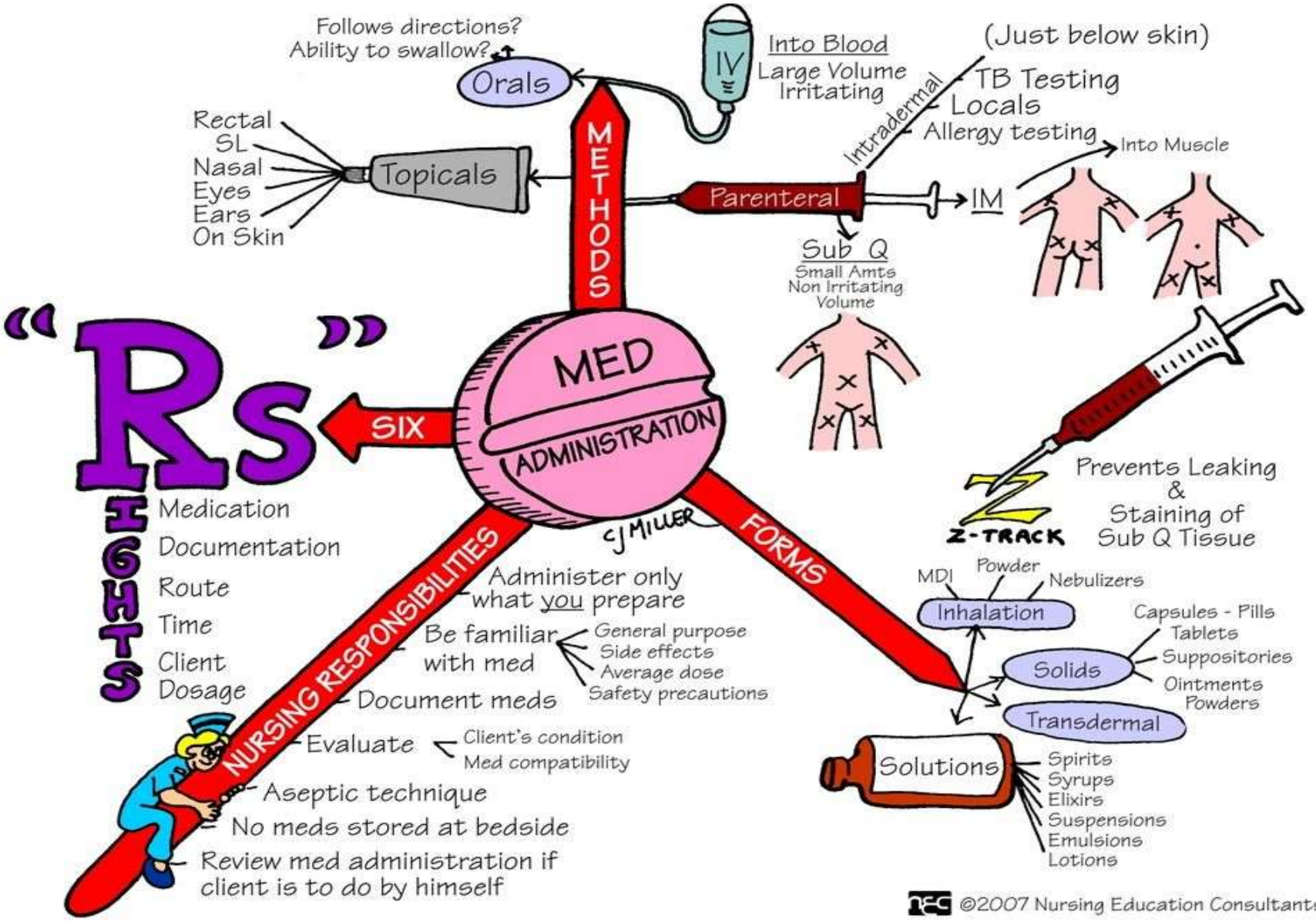
Drug calculation for nursing (Medication Administration)

Hayat abu nussera

Medication administration



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- Assess the 10 rights of medication



Summary of Product Characteristics

- The most important information regarding the drug is documented in the Summary of Product Characteristics – the SPC, also called the data sheet
- which is officially approved when the medicine is licensed for use.
- Its provide essential information to ensure that the drug or medicine is used correctly, effectively and safely

What an SPC contains:

1. Trade name of the medicinal product

The brand name of the drug or medicine .

Heparin (Mucous) Injection BP

2. Qualitative and quantitative composition

The generic or chemical names of the active ingredients and the amount of each active ingredient, e.g. amount per tablet, amount per volume of solution.

- **Heparin (Mucous) Injection BP 1000 Units/ml: each ml contains Heparin Sodium 1,000 IU.**

- **Heparin (Mucous) Injection BP 5000 Units/ml: each ml contains Heparin Sodium 5,000 IU.**

3. Pharmaceutical form

The form in which the medicine is presented, e.g.
tablets, suppositories,
ointment.

Solution for injection

4. Clinical particulars

How the medicine should be used; includes information for prescribers to ensure that patients are treated appropriately, taking into account the

patient's medical history, any co-existing diseases or conditions and other current treatments.

.

4.1 Therapeutic indications

The diseases or conditions that the medicine is licensed to treat.

- **For the treatment of thrombo-embolic disorders such as deep vein thrombosis, acute arterial embolism or thrombosis, thrombophlebitis, pulmonary embolism and fat embolism.**

- **For prophylaxis against deep vein thrombosis and thrombo-embolic events in susceptible patients.**

- **For the prevention of clotting in the extracorporeal circuit during haemodialysis.**

4.2 Posology and method of administration

Posology refers to the science of dosage, i.e. **how much of a medicine should be given to a patient.** The dose of the drug is given, including any changes in dose that may be necessary according to age or co-existing disease or condition, such as renal or hepatic impairment. Where relevant, information is also given on the timing of doses in relation to meals.

The maximum single dose, the maximum daily dose and the maximum dose for a course of treatment may also be given

- **Intravenous administration**

  5,000-10,000 IU every 4 hours or 500 IU/kg bodyweight daily as a continuous infusion in sodium chloride injection or dextrose injection. Doses should be individually adjusted according to coagulation tests.

- **Subcutaneous administration**

  The initial dose is 250 IU/kg bodyweight. Further doses should be given every 12 hours and individually adjusted according to coagulation tests.

4.3 Contra-indications

Anything about a patient's condition, medical history or current treatments that may indicate that this medicine should not be given.

Active major haemorrhage and risk factors for major haemorrhage. Generalised or local haemorrhagic tendency, including uncontrolled severe hypertension, severe liver insufficiency, active peptic ulcer, intracranial haemorrhage or injuries and operations on the central nervous system, eyes and ears, and in women with abortus imminens. This list is not exhaustive. Septic endocarditis.

4.4 Special warnings and special precautions for use

Any circumstances or condition where the drug should be used with particular care

Care should be taken when heparin is administered to patients with increased risk of bleeding complications, hypertension, renal or hepatic insufficiency. This list is not exhaustive.

.

4.5 Interactions with other medicaments and other forms of interaction

Any other medicines, or anything else that the patient is likely to take, which may react with the drug, for example, food or alcohol as well as other drugs.

- The anticoagulant effect of heparin may be enhanced by concomitant administration of other drugs affecting the coagulation system, such as those inhibiting platelet function (e.g. acetylsalicylic acid, other non-steroidal anti-inflammatory drugs (NSAIDs)

4.6 Pregnancy and lactation

Advice on the risks associated with using the drug at various stages of pregnancy and in fertile women, based upon the results of animal studies and any published observations in humans. If the drug or any of its metabolites is excreted in breast milk, the probability and nature of any adverse effects in the infant are described, and whether breast feeding should continue or not.

(Pregnancy)Heparin does not cross the placenta and can be used during all trimesters of pregnancy if clinically needed.

Breast-feeding **Heparin** is not excreted in human milk and can be used during breast-feeding.

4.7 Effects on ability to drive and use machines

Whether or not the medicine is likely to impair a patient's ability to drive or operate machinery and if so, the extent of the effect.

Heparin has no or negligible influence on the ability to drive or use machines.

4.8 Undesirable effects

A description of the adverse effects which may occur, including how likely they are to happen, how severe they may be and for how long they are likely to last.

The most frequently reported adverse reactions are haemorrhage and erythema.

4.9 Overdose

A description of the signs and symptoms of overdose, together with advice on how to treat.

Haemorrhage is the main complication of overdose.

5. Pharmacological properties

Information about how the medicine works and how it is handled by the body.

5.1 Pharmacodynamic properties

How the medicine achieves, or is believed to achieve, its therapeutic effect in the body.

Heparin is a naturally occurring anticoagulant which prevents the coagulation of blood *in-vivo* and *in-vitro*. It potentiates the inhibition of several activated coagulation factors, including thrombin and factor X.

5.2 Pharmacokinetic properties

Information on how the medicine is taken up, distributed in the body and then removed. Where appropriate, additional information may be included as to how the pharmacokinetics may change according to, for example, the patient's age or state of health.

The increase in clotting time provided by heparin becomes apparent immediately after administration and lasts for four to six hours after intravenous injection and for about eight hours after subcutaneous injection

5.3 Preclinical safety data

Describes the effects of the drug that were observed in studies before being used in humans, which could be of relevance to the prescriber in assessing the risks and benefits of treatment.

There are no preclinical data of relevance to the prescriber which are additional to that already included in other sections of the SPC.

6. Pharmaceutical particulars

Information on the medicine ingredients, storage and packaging.

6.1 List of excipients

The contents of the medicine apart from the active ingredients, such as
binding agents, solvents and flavourings are given.

Benzyl alcohol
Methylhydroxybenzoate (E218)
Propylhydroxybenzoate (E216)
Sodium citrate dihydrate
Sodium chloride
Water for injections

6.2 Incompatibilities

In addition to the information given under 'Interaction with other medicaments and other forms of interaction', any other medicines or materials that interact with the drug and with which it should therefore not be used or mixed.

Heparin has been reported to be incompatible in aqueous solution with certain substances, e.g. some antibiotics, hydrocortisone, phenothiazines, narcotic analgesics and some antihistamines.

6.3 Shelf life

The maximum length of time for which the medicine may be stored under the specified conditions stated and after which it should not be used.

3 years.

Once opened, the product may be stored for a maximum of 14 days below 25°C

6.4 Special precautions for storage

The conditions in which the medicine must be stored to avoid degradation, for example, by excessive temperatures or light.

Store below 25°C.

6.5 Nature and contents of container

Description of the packaging and any other materials included in the pack, such as desiccants, are given.

5 x 5 ml vials, 10 x 5 ml vials, 50 x 5 ml vials.

Not all pack sizes may be marketed

6.6 Instructions for use/handling

Instructions for the preparation or administration of the medicine in addition to those given under 'Posology and method of administration'.

7. Marketing authorization holder

The drug company holding the marketing authorization granted by the licensing authority.

- Heparin (Mucous) Injection BP 1000 Unit/ml PL 00043/0041R
- Heparin (Mucous) Injection BP 5000 Unit/ml PL 00043/0038R

8. Marketing authorization number

The licence number for the marketing authorization granted by the licensing authority.

LEO Laboratories Limited
Horizon
Honey Lane
Hurley
Maidenhead
Berkshire SL6 6RJ
UK

Medication Measurement Systems

A major type of medication measurement systems:

A. Metric system

B. Apothecary and household systems

C.Additional common medication measures

A. Metric system

1. The basic units of metric measures are the meter,

liter, and gram.

2. Meter measures length; liter measures volume;

gram measures mass

B. Apothecary and household systems

1. The apothecary and household systems are the oldest of the medication measurement systems.

2. Apothecary measures such as grain, dram, minim, and ounce are not commonly used in the clinical setting.

3. Commonly used household measures include drop, teaspoon, tablespoon, ounce, pint, and cup.

Additional common medication measures

1. Milliequivalent
    a. Milliequivalent is abbreviated mEq.
    b. The milliequivalent is an expression of the number of grams of a medication contained in 1 mL of a solution.
    c. For example, themeasure of serum potassium is given in milliequivalents.
2. Unit
    a. Unit measures a medication in terms of its action, not its physical weight.
    b. For example, penicillin, heparin sodium, and insulin are measured in units.

Common Pharmacologic Abbreviations

- G,gm gram
- gtt drop
- Kg kilogram
- L liter
- mcg microgram
- mEq milliequivalent
- Mg milligram
- ml milliliter
- U unit
- mu milliunit
- oz. Ounce
- qt quart
- Tbs, T tablespoon
- Tsp, t teaspoon

DRUG DOSAGE FORMS

- Cap capsule
- Elix elixir
- Liq liquid
- Sol solution
- Supp suppository
- Susp suspension
- Syr syrup
- Tab tablet
- oit ointment

ROUTES OF DRUG ADMINISTRATION

- IM intramuscular
- IV intravenous
- PO by mouth
- R, PR by rectum
- R right
- L left
- SC, SQ subcutaneous

TIMES OF DRUG ADMINISTRATION

- Ac before meals
- ad lib as desired
- Bid twice a day
- HS at bedtime
- pc after meals
- Prn as needed
- Q am, QM every morning
- QD, qd every day
- Qh every hour
- Q2h every 2 hours
- Q3h every 3 hours, and so on
- Qid four times a day
- Qod every other day
- STAT immediately
- Tid three times a day

COMMON INTRAVENOUS FLUIDS

- D5W – 5% Dextrose in water
- D5NS – 5% Dextrose in normal saline
- D5 ½NS – 5% Dexrose in ½ normal saline
- L.R. – Lactated Ringers

Remember 1 liter = 1000 ml

- c with
- C/O complains of
- D/C discontinue
- DX diagnosis
- HX history
- KVO keep vein open
- NKA no known allergies
- NKDA no known drug allergies
- NPO nothing by mouth
- R/T related to
- Rx treatment, prescription
- S without
- S/S signs/symptoms
- Sx symptoms
- TO telephone order
- VO verbal order

BASIC MATH REVIEW

- **1- Roman Numerals**
- I = 1
- V = 5
- X = 10
- L = 50
- C = 100
- D = 500
- M = 1000

- The basic form is to place the larger numerals to the left and add other numerals.

  XXXIII = 33 (30 + 3 = 33)

- There is an exception to the basic form.If smaller numeral precedes a larger numeral, the smaller should be subtracted from the larger.

  IX = 9 (1 - 10 = 9)

- 1. xvi =
- 2. CDXII =
- 3. XLVII =
- 4. XXi =
- 5. XLIV =
- 6. MCXX

Unit Conversion

- **<u>Metric Measurements of Weight</u>**
- Kilogram **kg** Approx. the weight of a litre of water
- Gram **g** One thousand grams to a kilogram
- Milligram **mg** One thousand mg to the gram
- Microgram **mcg** One million mcg to the gram
- Nanogram **ng** One thousand ng to the mcg

- **Metric Measurements of Liquids**
- Litre **l** Abbreviation is a lower-case L
- Millilitre **ml** One thousand millilitres to a litre

Metric system

Length	*Weight*	*Volume*
1 km = 1,000 m	1 kg = 1,000 g	1 kL = 1,000 L
1 m = .001 km	1 g = .001 kg	1 L = .001 kL
1 m = 100 cm	1 g = 100 cg	1 L = 100 cL
1 cm = .01 m	1 cg = .01 g	1 cL = .01 L
1 m = 1,000 mm	1 g = 1,000 mg	1 L = 1,000 mL
1 mm = .001 m	1 mg = .001 g	1 mL = .001 L

Household System

- This system of measure is not as accurate as the metric or apothecary systems.

- The units of volume include drop (gtts), teaspoon (tsp or t.), tablespoon (tbsp. or T) and ounces (oz.).

Household system

Measurement Unit	Equivalent within System	Metric Equivalent
1 tsp (teaspoonful)		5 mL
1 tbsp (tablespoonful)	3 tsp	15 mL
1 fl oz (fluid ounce)	2 tbsp	30 mL (29.57 mL)*
1 cup	8 fl oz	240 mL
1 pt (pint)	2 cups	480 mL*
1 qt (quart)	2 pt	960 mL
1 gal (gallon)	4 qt	3840 mL

Table 5.7
Household Measure: Weight

Measurement Unit	Equivalent within System	Metric Equivalent
1 oz (ounce)		30 g
1 lb (pound)	16 oz	454 g
2.2 lb		1 kg

Edited by Dr. Ryan Lambert-Bellacov, chiropractor for Back in the Game in West Linn, OR

Data Types in Statistics

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1. Introduction to Data Types in Statistics and their Importance

2. Qualitative vs Quantitative Data

3. Qualitative Data

- Nominal Data
- Ordinal Data

4. Quantitative Data

- Discrete Data
- Continuous Data
- Interval Data
- Ratio Data

Introduction to Data Types in Statistics

In Statistics, Data Types play a very crucial and important role, which needs to be understood, to apply statistical measurements correctly to your data so that we can correctly conclude certain assumptions about the data.

Similarly, we need to know which data analysis and its type you are working on to select the correct perception technique since different data types are considered as an approach to arrange various types of variables.

While doing **Exploratory Data Analysis (EDA)** in a general data science project, it becomes crucial to have a good understanding of the different data types since we can use certain statistical measurements only for specific data types.

It is also known as the **Measurement Scale**.

While dealing with any of the data types, we also need to know which visualization method fits the particular data type.

We can think of data types as a way to categorize different types of variables.

Quantitative vs Qualitative Data

Quantitative Data

1. These types of data seem to be the easiest to explain. It tries to find the answers to questions such as

- “how many,
- “how much” and

- "how often"

2. It can be expressed as a number, so it can be quantified. In simple words, it can be measured by numerical variables.

3. These are easily open for statistical manipulation and can be represented by a wide variety of statistical types of graphs and charts like **line charts**, **bar graphs**, **scatter plots**, etc.

Examples of quantitative data:

- Scores of tests and exams e.g. 74, 67, 98, etc.
- The weight of a person.
- The temperature in a room.

There are 2 general types of quantitative data:

- Discrete data
- Continuous data

Qualitative Data

1. Qualitative data can't be expressed as a number, so it can't be measured. It mainly consists of words, pictures, and symbols, but not numbers.

2. It is also known as **Categorical Data** as the information can be sorted by category, not by number.

3. These can answer the questions like:

- "how this has happened", or
- "why this has happened".

Examples of qualitative data:

- Colors e.g. the color of the sea
- Popular holiday destinations such as Switzerland, New Zealand, South Africa, etc.
- Ethnicity such as American Indian, Asian, etc.

In general, there are 2 types of qualitative data:

- Nominal data
- Ordinal data.

Qualitative Data

Nominal Data

1. The lowest level of data. This data type is used just for labeling variables, without having any quantitative value. Here, the term 'nominal' comes from the Latin word **"nomen"** which means **'name'**.

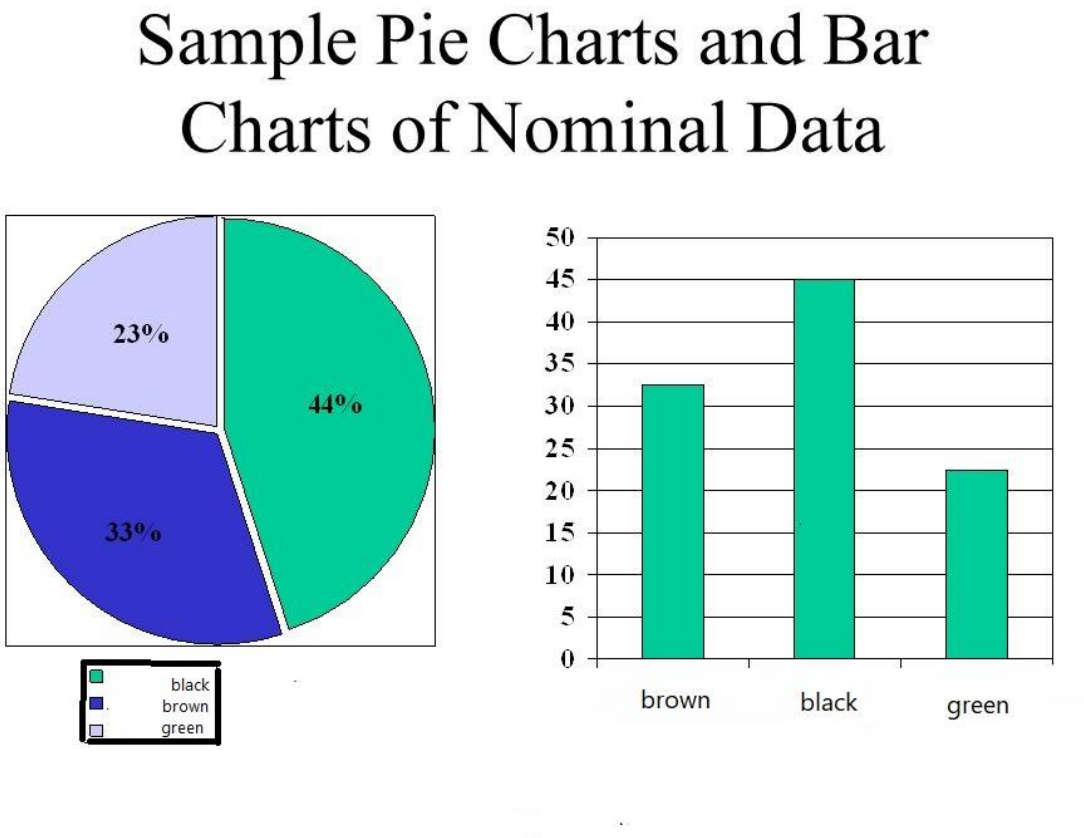
2. It just names a thing without applying for any particular order. The nominal data sometimes referred to as **"labels".**

Examples of Nominal Data:

- Gender (Women, Men)
- Hair color (Blonde, Brown, Brunette, Red, etc.)
- Marital status (Married, Single, Widowed)

As you can observe from the examples there is no intrinsic ordering to the variables.

Eye color is a nominal variable having a few levels or categories such as Blue, Green, Brown, etc and there is no possible way to order these categories in a rank-wise manner i.e, from highest to lowest or vice-versa.



Ordinal Data

1. The crucial difference from nominal types of data is that Ordinal Data shows where a number is present in a particular order.

2. This type of data is placed into some kind of order by their position on a scale. Ordinal data may indicate superiority.

3. We cannot do arithmetic operations with ordinal data because they only show the sequence.

4. Ordinal variables are considered as "in-between" qualitative and quantitative variables.

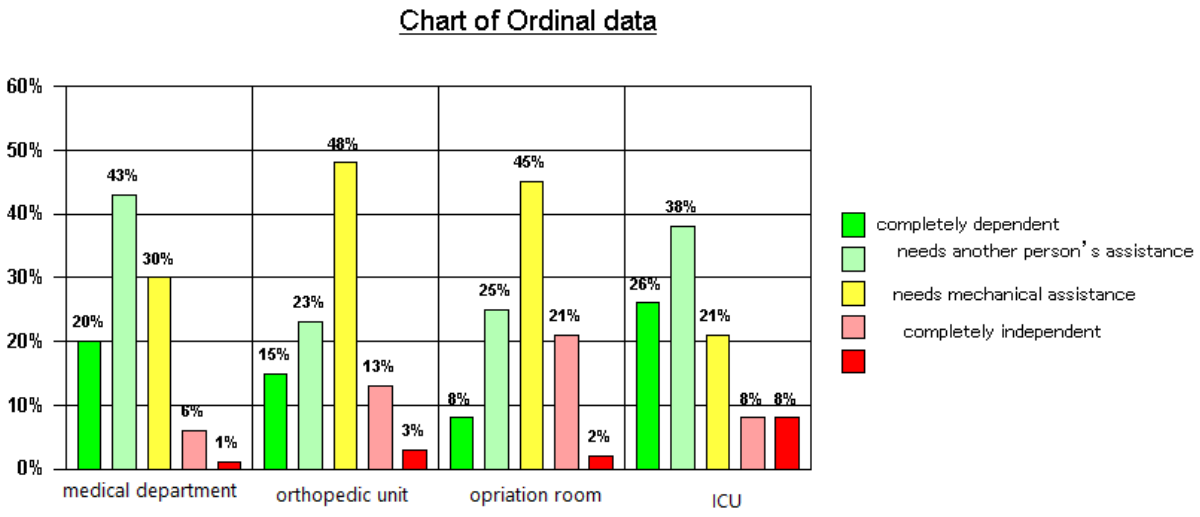
5. In simple words, we can understand the ordinal data as qualitative data for which the values are ordered.

6. In comparison with nominal data, the second one is qualitative data for which the values cannot be placed in an order.

7. Based on the relative position, we can also assign numbers to ordinal data. But we cannot do math with those numbers. **For example,** "first, second, third…etc."

Examples of Ordinal Data:

- Ranking of users in a competition: The first, second, and third, etc.
- Rating of a product taken by the company on a scale of 1-10.
- Economic status: low, medium, and high.
- For example, consider this ordinal coding scheme for measuring ability to perform activities of daily living (ADL): 1 = completely dependent, 2 = needs another person's assistance, 3 = needs mechanical assistance, and 4 = completely independent. The numbers signify incremental ability to perform ADL independently. Ordinal measurement does not, however, tell us how much greater one level is than another. For example, we do not know if being completely independent is twice as good as needing mechanical assistance.



Quantitative Data

Discrete Data

1. It shows the count that involves only integers and we cannot subdivide the discrete values into parts.

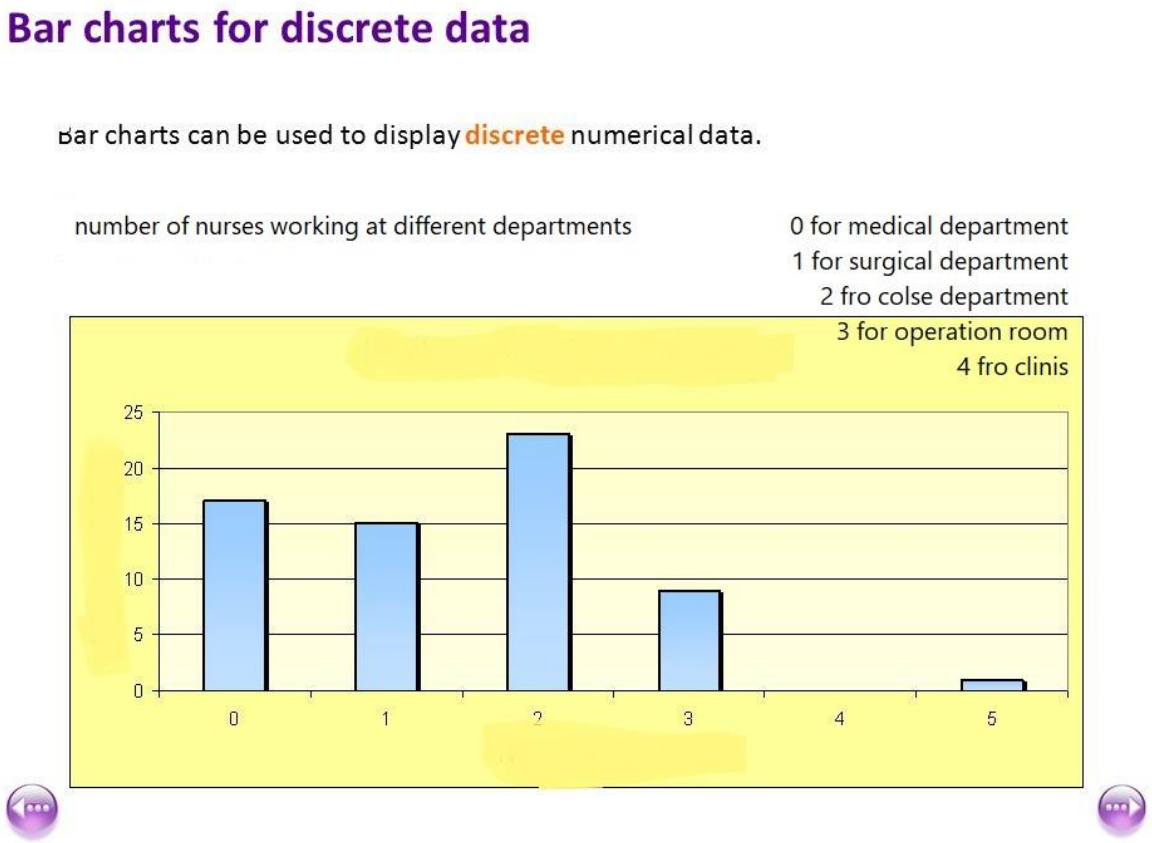
For Example, the number of students in a class is an example of discrete data since we can count whole individuals but can't count like 2.5, 3.75, kids.

2. In simple words, discrete data can take only certain values and the data variables cannot be divided into smaller parts.

3. It has a limited number of possible values **e.g. days of the month.**

Examples of discrete data:

- The number of students in a class.
- The number of workers in a company.
- The number of test questions you answered correctly.



Continuous Data

1. It represents the information that could be meaningfully divided into its finer levels. It can be measured on a scale or continuum and can have almost any numeric value.

For Example, We can measure our height at very precise scales in different units such as meters, centimeters, millimeters, etc.

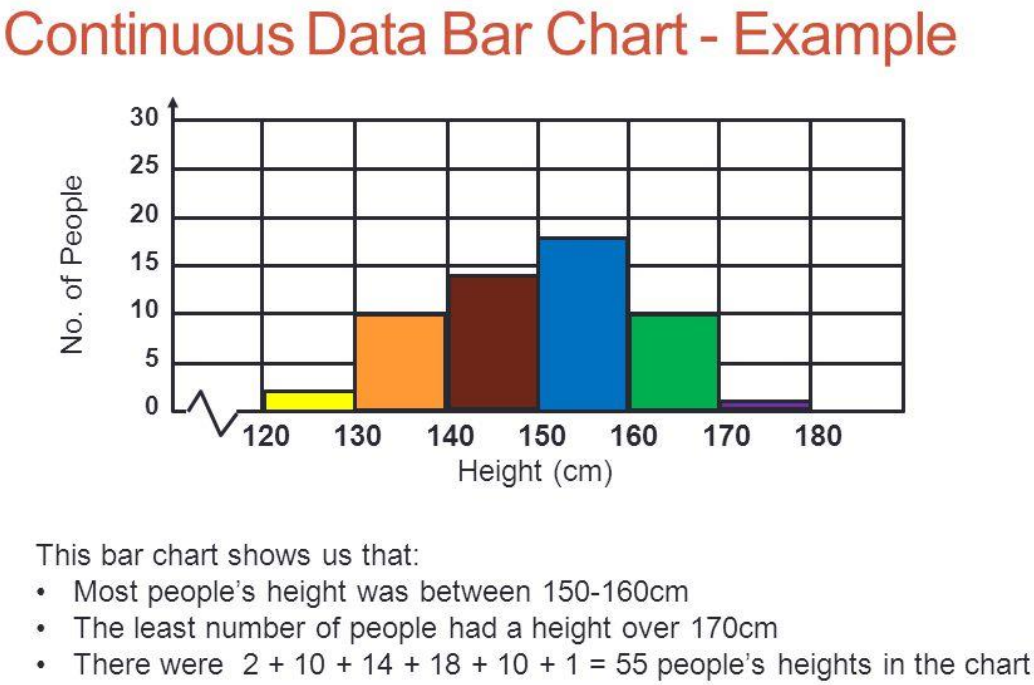
2. The key difference between continuous and discrete types of data is that in the former, we can record continuous data at so many different measurements such as width, temperature, time, etc.

3. The continuous variables can take any value between two numbers. **For Example,** between the range of 60 and 82 inches, there are millions of possible heights such as 62.04762 inches, 79.948376 inches, etc.

4. A good great rule for defining if data is continuous or discrete is that if the point of measurement can be reduced in half and still make sense, the data is continuous.

Examples of continuous data:

- The amount of time required to complete a project.
- The height of children.
- The speed of cars.



Interval Data

1. These data types are measurable and ordered with the nearest items but have no meaningful zero.

Let's understand the meaning of "Interval Scale":

In the Interval scale, the term 'Interval' signifies space in between, which is a significant thing to recall as interval scales not only educate us about the order but in addition, give information about the value between every item.

2. Fundamentally, we can show interval data in the same way as ratio data, but the thing that we have to note is their characterized zero points.

3. Hence, with the help of interval data, we can easily correlate the degrees of the data and also add or subtract the values.

4. There are some descriptive statistics that we can calculate for interval data such as :

- Central measures of tendency (mean, median, mode)
- Range (minimum, maximum)
- Spread (percentiles, interquartile range, and standard deviation).

These are not the only statistical things to be calculated, but we can calculate more things also.

Examples of Interval data:

- Temperature (°C or F, but not Kelvin)
- Dates (1055, 1297, 1976, etc.)
- Time Gap on a 12-hour clock (6 am, 6 pm)

Ratio Data

1. These data are also in the ordered units that have the same difference.

2. Ratio values are the same as interval values, but the only difference is that Ratio data do have an absolute zero. **For Example,** height, weight, length, etc.

3. These are measured and ordered with equidistant items with a meaningful zero and never be negative like interval data.

Let's understand this with an outstanding example- **Measurement of heights**.

Height can be measured in units like centimeters, inches, meters, or feet and it is not possible to have a negative value of height.

4. It enlightens us regarding the order for variables, the contrasts among them, and they have absolutely zero.

5. Ratio data is fundamentally the same as interval data, aside from zero means none.

6. The descriptive statistics which we can calculate for ratio data are the same as interval data such as :

- Central measures of tendency (mean, median, mode)
- Range (minimum, maximum)

- Spread (percentiles, interquartile range, and standard deviation).

Example of Ratio data:

- Age (from 0 years to 100+)
- Temperature (in Kelvin, but not °C or F)
- Time interval (measured with a stop-watch or similar)

For the above examples of ratio data, we see that there is an actual and meaningful zero-point like the age of a person, absolute zero, distance calculated from a specified point or time all have real zeros.

NOTE:

If we picked the zero-point of the scale subjectively, then at that point the data can't be ratio data and should be interval data.

Statistical Analysis of Quantitative Data

DESCRIPTIVE STATISTICS

Statistical procedures enable researchers to organize and interpret numeric information. Statistics are either descriptive or inferential. **Descriptive statistics** are used to synthesize and describe data. When indexes such as averages and percentages are calculated with data from a population, they are **parameters.** A descriptive index from a sample is a **statistic.** Most research questions are about parameters; researchers calculate statistics to estimate parameters and use **inferential statistics** to make inferences about the population. A set of data for a variable can be described in terms of three characteristics: the shape of the distribution of values, central tendency, and variability. Frequency Distributions Data that are not organized are overwhelming. Without some structure, even broad trends are hard to discern. Consider the 60 numbers in Table 12.1. Let us assume that these numbers are the scores of 60 preoperative patients on a six-item scale of anxiety. Visual inspection of these numbers provides little insight on patients′ anxiety.

TABLE 12.1 Patients' Anxiety Scores

22	27	25	19	24	25	23	29	24	20
26	16	20	26	17	22	24	18	26	28
15	24	23	22	21	24	20	25	18	27
24	23	16	25	30	29	27	21	23	24
26	18	30	21	17	25	22	24	29	28
20	25	26	24	23	19	27	28	25	26

Frequency distributions impose order on numeric data. A **frequency distribution** is a systematic arrangement of values from lowest to highest, together with a count or percentage of how many times each value occurred.

A frequency distribution for the 60 anxiety scores makes it easy to see the highest and lowest scores, the most common score, where the scores clustered, and how many patients were in the sample (total sample size is designated as ***N*** in research reports). None of this was apparent before the data were organized. (table 12.2).

Frequency data can be displayed graphically in a *frequency polygon* (Fig. 12.1). In such graphs, scores typically are on the horizontal line, with the lowest value on the left, and frequency counts or percentages are on the vertical line. Data distributions can be described by their shapes. **Symmetric distribution** occurs if, when folded over, the two halves of a frequency polygon would be superimposed (Fig. 12.2). In an asymmetric or **skewed**

distribution, the peak is off center, and one tail is longer than the other. When the longer tail points to the right, the distribution has a **positive skew,** as in the first graph of Figure 12.3. Personal income is a positively skewed attribute. Most people have moderate incomes, with very high incomes at the right end of the distribution. If the longer tail points to the left, the distribution has a **negative skew,** as in the second graph in Figure 12.3. Age at death is negatively skewed: the bulk of people are at the far right end of the distribution, with relatively few people dying at an early

TABLE 12.2 Frequency Distribution of Patients' A

Score	Frequency	Percentage (%)
15	1	1.7
16	2	3.3
17	2	3.3
18	3	5.0
19	2	3.3
20	4	6.7
21	3	5.0
22	4	6.7
23	5	8.3
24	9	15.0
25	7	11.7
26	6	10.0
27	4	6.7
28	3	5.0
29	3	5.0
30	2	3.3
	$N = 60$	100.0%

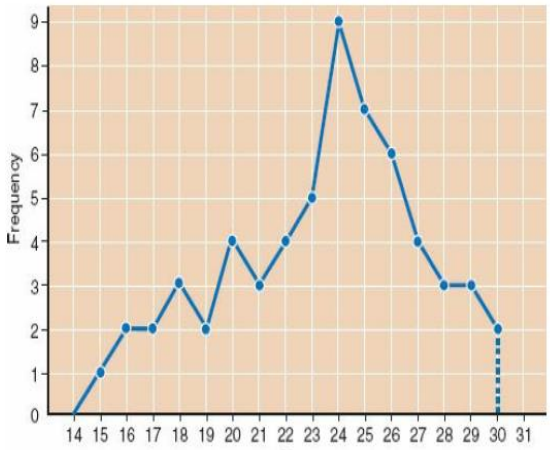


FIGURE 12.1 • Frequency polygon of patients' anxiety scores.

Central Tendency

Frequency distributions help to clarify patterns, but often a pattern is less useful than an overall summary. Researchers usually ask a question such as, "What is the *average* oxygen consumption of myocardial infarction patients during bathing?" Such a question seeks a single number to represent a distribution of values. Indexes of "typicalness" are called measures of **central tendency.** Researchers avoid using the term *average* because there are three indexes of central tendency: the mode, the median, and the mean.

• **Mode:** The mode is the number that occurs most frequently in a distribution. In the following distribution, the mode is 53:

50 51 51 52 53 53 53 53 54 55 56

The value of 53 occurred four times, a higher frequency than for other numbers. The mode of the patients′ anxiety scores in Table 12.2 was 24. The mode, which identifies the most " popular" value, is used most often to describe high-frequency values for nominal measures.

• **Median:** The median is the point in a distribution that divides scores in half. Consider the following set of values:

2 2 3 3 4 5 6 7 8 9

The value that divides the cases in half is midway between 4 and 5, and thus 4.5 is the median. The median for the anxiety scores in Table 12.2 is 24, the same as the mode. The median does not take into account individual values and so is insensitive to extremes. In the above set of numbers, if the value of 9 were changed to 99, the median would remain 4.5. Because of this property, the median is the preferred index to describe a highly skewed distribution. In research articles, the median may be abbreviated as *Md* or *Mdn*.

• **Mean:** The mean equals the sum of all values divided by the number of participants— what people refer to as the average. The mean of the patients′ anxiety scores is 23.4 (1,405 ÷ 60). As another example, here are the weights of eight people:

85 109 120 135 158 177 181 195

In this example, the mean is 145. Unlike the median, the mean is affected by the value of every score. If we exchanged the 195-lb person for one weighing 275 lb, the mean would increase from 145 to 155 lb. In research articles, the mean is often symbolized as *M* or (e.g., = 145). For interval-level or ratio-level measurements, the mean is usually the statistic reported. Of the three indexes, the mean is the most stable: if repeated samples were drawn from a population, the means would fluctuate less than the modes or medians. Because of its stability, the mean usually is the best estimate of a population′s central tendency. When a distribution is highly skewed, however, the median is preferred. For example, the median is a

better central tendency measure of family income than the mean because income is positively skewed.

Variability

Two distributions with identical means could differ with respect to shape (e.g., how skewed they are) and how spread out the data are (i.e., how different people are from one another on an attribute). This section describes the **variability** of distributions.

• **Range:** The range is the highest score minus the lowest score in a distribution. In our anxiety score example, the range is 15 (30 − 15). In the distributions in Figure 12.4, the range for school A is about 80 (140 − 60), whereas the range for school B is about 50 (125 − 75). The chief virtue of the range is ease of computation. Because it is based on only two scores, however, the range is unstable: from sample to sample drawn from a population, the range can fluctuate greatly. The range is used largely as a gross descriptive index.

• **Standard deviation:** The most widely used variability index is the standard deviation. Like the mean, the standard deviation is calculated based on every value in a distribution. The standard deviation summarizes the *average* amount of deviation of values from the mean.* In the anxiety scale example, the standard deviation is 3.725.** In research reports, the standard deviation may be abbreviated as *s* or *SD*.

A standard deviation (*SD*) is more difficult to interpret than the range. For the *SD* of anxiety scores, you might ask, 3.725 *what?* What does the number mean? We can answer these questions from several angles. First, the *SD* is an index of how variable scores in a distribution are and so if (for example) male and female patients had means of 23 on the anxiety scale, but their *SD*s were 7 and 3, respectively, we would know that females were more homogeneous (i.e., their scores were more similar to one another).

The *SD* represents the *average* of deviations from the mean. The mean tells us the best value for summarizing an entire distribution, and an *SD* tells us how much, on average, the scores deviate from the mean. In the anxiety scale example, scores deviated by an average of just under 4 points. A standard deviation can be interpreted as our degree of error when we use a mean to describe an entire sample.

Bivariate Descriptive Statistics

So far, our discussion has focused on *univariate* (one-variable) *descriptive statistics.* A mean or *SD* describe one variable at a time. *Bivariate* (twovariable) *descriptive statistics* describe relationships between two variables.

Crosstabulations

A **crosstabs table** (or *contingency table*) is a two-dimensional frequency distribution in which the frequencies of two variables are **crosstabulated**. Suppose we had data on patients′ sex and whether they were nonsmokers, light smokers (less than one pack of cigarettes a day), or heavy smokers (one or more packs a day). The question is whether men smoke more heavily than women, or vice versa (i.e., whether there is a *relationship* between smoking and sex). Fictitious data for this example are shown in Table 12.4.

TABLE 12.4 Crosstabs Table for Relationship Between Sex and Smoking Status

	Sex					
	Women		Men		Total	
Smoking Status	***n***	**%**	***n***	**%**	***n***	**%**
Nonsmoker	10	45.4	6	27.3	16	36.4
Light smoker	8	36.4	8	36.4	16	36.4
Heavy smoker	4	18.2	8	36.4	12	27.3
TOTAL	22	100.0	22	100.0	44	100.0

Correlation

Relationships between two variables can be described by **correlation** methods. The correlation question is: To what extent are two variables relatedto each other? For example, to what degree are anxiety scores and blood pressure values related? This question can be answered by calculating a **correlation coefficient,** which describes *intensity* and *direction* of a relationship.

Two variables that are related are height and weight: tall people tend to weigh more than short people. The relationship between height and weight would be a *perfect relationship* if the tallest person in a population was the heaviest, the second tallest person was the second

heaviest, and so on. A correlation coefficient indicates how "perfect" a relationship is.

Possible values for a correlation coefficient range from −1.00 through .00 to +1.00. If height and weight were perfectly correlated, the correlation coefficient would be 1.00 (the actual correlation coefficient is in the vicinity of .50 to .60 for a general population). Height and weight have a **positive relationship** because greater height tends to be associated with greater weight. When two variables are unrelated, the correlation coefficient is zero. One might anticipate that women's shoe size is unrelated to their intelligence.

Women with large feet are as likely to perform well on IQ tests as those with small feet. The correlation coefficient summarizing such a relationship would be in the vicinity of .00.

Correlation coefficients between .00 and −1.00 express a **negative** (*inverse*) **relationship.** When two variables are inversely related, increments in one variable are associated with decrements in the second. For example, there is a negative correlation between depression and self-esteem. This means that, on average, people with *high* self-esteem tend to be *low* on depression. If the relationship were perfect (i.e., if the person with the highest self-esteem score had the lowest depression score and so on), then the correlation coefficient would be −1.00. In actuality, the relationship between depression and self-esteem is moderate—usually in the vicinity of −.40 or −.50. Note that the higher the *absolute value* of the coefficient (i.e., the value disregarding the sign), the stronger the relationship. A correlation of −.80, for instance, is much stronger than a correlation of +.20. The most commonly used correlation index is **Pearson's *r*** (the *product– moment correlation coefficient*), which is computed with interval or ratio measures. There are no fixed guidelines on what should be interpreted as strong or weak relationships, because it depends on the variables. If we measured patients' body temperature orally and rectally, an ***r*** of .70 between the two measurements would be low. For most psychosocial variables (e.g., stress and depression), however, an *r* of .70 would be high. Perfect correlations (+1.00 and −1.00) are rare.

BIVARIATE STATISTICAL TESTS

Researchers use a variety of statistical tests to make inferences about their hypotheses. Several frequently used bivariate tests are briefly described and illustrated.

t-Tests

Researchers frequently compare two groups of people on an outcome. A parametric test for testing the significance of differences in two group means is called a ***t*-test**. Suppose we wanted to test the effect of early discharge of maternity patients on perceived maternal competence. We administer a scale of perceived maternal competence at discharge to 20 primiparas who had a vaginal delivery: 10 who remained in the hospital 25 to 48 hours (regular discharge group) and 10 who were discharged 24 hours or less after delivery (early discharge group). Data for this example are presented in Table 12.8. Mean scores for these two groups are 25.0 and 19.0, respectively. Are these differences *real* (i.e., would they exist in the population of early-discharge and later-discharge mothers?), or do group differences reflect chance fluctuations? The 20 scores vary from one mother to another. Some variation reflects individual differences in perceived maternal competence. Some variation might reflect the scale′s low reliability, some could result from participants′ moods on a particular day, and so forth. The research question is: Can a significant portion of the variation be attributed to the independent variable— time of hospital discharge? The *t*-test allows us to make inferences about this question objectively.

TABLE 12.8 Fictitious Data for *t*-Test Example: Scores on a Perceived Maternal Competence scale

Early-Discharge Mothers	Regular-Discharge Mothers
30	23
27	17
25	22
20	18
24	20
32	26
17	16
18	13
28	21
29	14
Mean = 19.0	Mean = 25.0
$t = 2.86$, $df = 18$, $p = .011$	

The formula for calculating the *t* statistic uses group means, variability, and sample size. The computed value of *t* for the data in Table 12.8 is 2.86. Degrees of freedom in this example are equal to the total sample size minus 2 ($df = 20 - 2 = 18$). For an α level of .05, the theoretical cutoff value for *t* with 18 degrees of freedom is 2.10. *This value establishes an upper limit to what is probable if the null hypothesis is true.* Thus, the calculated *t* of 2.86, which is larger than the theoretical value of *t*, is improbable (i.e., statistically significant).

Analysis of Variance

Analysis of variance (ANOVA) is used to test mean group differences of three or more groups. ANOVA sorts out the variability of an outcome variable into two components: variability due to the independent variable (e.g., experimental group status) and variability due to all other sources (e.g., individual differences, measurement error). Variation *between* groups is contrasted with variation *within* groups to yield an ***F* ratio** statistic. Suppose we were comparing the effectiveness of interventions to help people stop smoking. One group of smokers receives nurse counseling (groupA), a second group is treated by a nicotine patch (group B), and a control group gets no special treatment (group C). The outcome is 1-day cigarette consumption measured 1 month after the intervention. Thirty smokers who wish to quit smoking are randomly assigned to one of the three groups. The null hypothesis is that the population means for post-treatment cigarette smoking is the same for all three groups, and the research hypothesis is inequality of means. Table 12.9 presents fictitious data for the 30 participants. The mean numbers of post-treatment cigarettes consumed are 16.6, 19.2, and 34.0 for groups A, B, and C, respectively. These means are different, but are they significantly different—or do differences reflect random fluctuations?

TABLE 12.9 Fictitious Data for ANOVA Example: Number Of Cigarettes Smoked In 1 Day, Post-Treatment

Group A Nurse Counseling		Group B Nicotine Patch		Group C Untreated Controls	
28	19	0	27	33	35
0	24	31	0	54	0
17	0	26	3	19	43
20	21	30	24	40	39
35	2	24	27	41	36
$Mean_A = 16.6$		$Mean_B = 19.2$		$Mean_C = 34.0$	

$F = 4.98$, $df = 2, 27$, $p = .01$

An ANOVA applied to these data yields an *F* ratio of 4.98. For a α .05 and *df* = 2 and 27 (2 *df* between groups and 27 *df* within groups), the theoretical *F* value is 3.35. Because our obtained *F*-value of 4.98 exceeds 3.35, we reject the null hypothesis that the population means are equal. The *actual* probability, as calculated by a computer, is .014. In only 14 samples out of 1,000 would differences this great be obtained by chance alone.

Chi-Squared Test

The **chi-squared** (χ2) **test** is used to test hypotheses about the proportion of cases in different categories, as in a crosstabulation. For example, suppose we were studying the effect of nursing instruction on patients′ compliance with self-medication. Nurses implement a new instructional strategy with 50 patients, while 50 control group patients get usual care. The research hypothesis is that a higher proportion of people in the treatment than in the control condition will be compliant. Some fictitious data for this example are presented in Table 12.10, which shows that 60% of those in the experimental group were compliant, compared to 40% in the control group. But is this 20 percentage point difference statistically significant—that is, likely to be ″real″?

TABLE 12.10 Observed Frequencies for Chi-Squared Example: Rates of Compliance with Medications

Patient Compliance	Group				Total
	Experimental		Control		
	n	%	*n*	%	*n*
Compliant	30	60.0	20	40.0	50
Noncompliant	20	40.0	30	60.0	50
TOTAL	50	100.0	50	100.0	100

$\chi^2 = 4.0$, $df = 1$, $p = .046$

The chi-squared statistic is computed by summing differences between the *observed frequencies* in each cell (such as those in Table 12.10) and the *expected frequencies*—those that would be expected if there were *no* relationship between the variables. The value of the chi-squared statistic here is 4.00, which we can compare with the value from a theoretical chi-squared distribution. In this example, the theoretical value that must be exceeded to establish significance at the .05 level is 3.84. The obtained value of 4.00 is larger than would be expected by chance (the actual $p = .046$). We can conclude that a significantly larger proportion of experimental patients than control patients were compliant.

Correlation Coefficients

Pearson′s *r* is both descriptive and inferential. As a descriptive statistic, *r* summarizes the magnitude and direction of a relationship between two variables. As an inferential statistic, *r* tests hypotheses about population correlations; the null hypothesis is that there is no relationship between two variables, that is, that the population $r = .00$. Suppose we were studying the relationship between patients′ self-reported level of stress (higher scores indicate more stress) and the pH level of their saliva. With a sample of 50 patients, we find that $r = -.29$. This value indicates a tendency for people with high stress to have lower pH levels than those with low stress. But we need to ask whether this finding can be generalized to the population. Does the *r* of $-.29$ reflect a random fluctuation, observed only in this sample, or is the relationship significant? Degrees of freedom for correlation coefficients equal *N* minus 2, which is 48 in this example. The theoretical value for *r* with $df = 48$ and a α .05 is .28. Because the absolute value of the calculated *r* is .29, the null hypothesis is rejected.

There is a modest but significant relationship between patients′ stress level and the acidity of their saliva. CIs can be constructed around Pearson *rs*. In our example, the 95% CI around the *r* of .29 for stress levels and saliva pH, with a sample of 50 subjects, is .01 and .53. Because the upper confidence limit is less than .00, the correlation in this example was statistically significant at the .05 level (but note that the range of possible values for the population *r* is very large because of the small sample size).

TABLE 12.11 Guide to Major Bivariate Statistical Tests

Name	Test Statistic	Purpose	Measurement Level	
			Independent Variable	Dependent Variable
t-test for independent groups	*t*	To test the difference between the means of two independent groups (e.g., experimental vs. control, men vs. women)	Nominal	Interval, Ratio
t-test for paired groups	*t*	To test the difference between the means of a paired group (e.g., pretest vs. posttest for same people)	Nominal	Interval, Ratio
Analysis of variance (ANOVA)	*F*	To test the difference among means of 3+ independent groups or means for 2+ independent variables	Nominal	Interval, Ratio
Repeated measures ANOVA	*F*	To test the difference among means of 3+ related groups, e.g., the same group over time, or to compare 2+ groups over time	Nominal	Interval, Ratio
Pearson's correlation coefficient	*r*	To test the existence of a relationship between two variables	Interval, Ratio	Interval, Ratio
Chi-squared test	χ^2	To test the difference in proportions in 2+ independent groups	Nominal (or ordinal, few categories)	Nominal (or ordinal, few categories)

PER CENT AND PERCENTAGES

OBJECTIVES

At the end of this chapter, you should be familiar with the following

- Percent and percentages
- Converting fractions to percentages and vice versa
- Converting decimals to percentages and vice versa
- Calculations involving percentages
- How to find the percentage of a number
- How to find what percentage one number is of another
- Drug calculations involving percentages
- How to use the percentage key on your calculator

CONVERTING FRACTIONS TO PERCENTAGES AND VICE VERSA

To convert a fraction to a percentage, you simply **multiply** by 100, e.g.

$$\frac{2}{5} = \left(\frac{2}{5} \times 100\right)\% = 40\%$$

Conversely, to convert a percentage to a fraction, **divide** by 100, i.e.

$$40\% = \frac{40}{100} = \frac{4}{10} = \frac{2}{5}$$

Remember, to divide by 100, you move the decimal point **two** places to the **left**. So, to convert fractions and decimals to percentages or vice versa, you simply multiply or divide by 100. Thus:

$$25\% = \frac{25}{100} = \frac{1}{4} = 0.25 \quad \text{(a quarter)}$$
$$33\% = \frac{33}{100} = \frac{1}{3} = 0.33 \quad \text{(approx. a third)}$$
$$50\% = \frac{50}{100} = \frac{1}{2} = 0.5 \quad \text{(a half)}$$
$$66\% = \frac{66}{100} = \frac{2}{3} = 0.66 \quad \text{(approx. two-thirds)}$$
$$75\% = \frac{75}{100} = \frac{3}{4} = 0.75 \quad \text{(three-quarters)}$$

The next step is to look at how to find the percentage of an amount. The Worked Example on the next page shows how this is done.

CALCULATIONS INVOLVING PERCENTAGES

The first type of calculation we are going to look at is how to find the percentage of a given quantity or number.

WORKED EXAMPLE

How much is 28% of 250?

There are several ways of solving this type of problem.

METHOD 1: WORKING IN PERCENTAGES

With this method you are working in percentages.

STEP ONE

When doing percentage calculations, the number or quantity you want to find the percentage of, is always equal to 100%. In this example, 250 is equal to 100% and you want to find out how much 28% is. So:

250 = 100%

(Thus you are converting the number to a percentage.)

STEP TWO

Calculate how much is equal to 1%, i.e. divide by 100 (you are using the

'ONE unit' rule – see Chapter 1 'First principles' for an explanation):

$$1\% = \frac{250}{100}$$

STEP THREE

Multiply by the percentage required (28%):

$$28\% = \frac{250}{100} \times 28 = 70$$

ANSWER: 28% of 250 is 70.

METHOD 2: WORKING IN FRACTIONS

In this method you are trying to find the fraction of the whole.

STEP ONE

First convert the percentage to a fraction, i.e. divide by 100:

$$28\% = \frac{28}{100}$$

STEP TWO

Multiply by the original number (250) to find how much the fraction is of the whole:

$$\frac{28}{100} \times 250 = 70$$

Thus you are finding out how much the fraction is of the original

number.

ANSWER: 28% of 250 is 70.

In both methods used, it can be seen that the number is always divided by 100; thus a simple formula can be devised. If we call the original number **base**; the amount we want to find out (i.e. the answer) the **amount**; and the percentage required, **per cent**, then we can write this as formula:

$$\text{amount} = \frac{\text{base}}{100} \times \text{per cent}$$

In this example:

$$\text{base} = 250$$
$$\text{per cent} = 28\%$$

Substituting the numbers in the formula:

$$\frac{250}{100} \times 28 = 70$$

ANSWER: 28% of 250 is 70.

WORKED EXAMPLE

What percentage is 630 of 9,000?

In this case, it is best to work in percentages since that is what you want to find.

STEP ONE

Once again, the number or quantity you want to find the percentage of is always equal to 100%.

In this example, 9,000 would be equal to 100% and you want to find out the percentage that is 630. So:

$$9{,}000 = 100\%$$

(Thus you are converting the number to a percentage.)

STEP TWO

Calculate the percentage that is 1, i.e. divide by 9,000 using the 'ONE unit' rule:

$$1 = \frac{100}{9{,}000}\%$$

STEP THREE

Multiply by the number you wish to find the percentage of, i.e. the smaller number (630):

$$\frac{100}{9{,}000} \times 630 = 7\%$$

ANSWER: 630 is 7% of 9,000.

Once again, it can be seen that you multiply by 100 and using the same terms as before (**base** would equal the larger number and **amount** the smaller number), a simple formula can be devised:

$$\text{per cent} = \frac{100}{\text{base}} \times \text{amount} \quad \textit{or} \quad \frac{\text{amount}}{\text{base}} \times 100$$

PROBLEMS

Work out the following:

Question 1 30% of 3,090

Question 2 84% of 42,825

Question 3 56.25% of 800

Question 4 60% of 80.6

Question 5 17.5% of 285.76

DRUG CALCULATIONS INVOLVING PERCENTAGES

The principles illustrated here can easily be applied to drug calculations. As before, it is unlikely that you will need to find the percentage of something, but these calculations are included here in order to gain an understanding of per cent and percentages, especially where drugs are concerned. Always, convert everything to the same units before doing the calculation.

WORKED EXAMPLE

What volume (in mL) is 60% of 1.25 litres?

Work in the same units – in this case, work in millilitres as these are the units required in the answer:

To convert to milliliters, multiply by 1,000

$$1.25 \text{ litres} = (1.25 \times 1{,}0000)\text{mL} = 1{,}250\text{mL}$$

Remember the number or quantity you want to find the percentage of is always equal to 100%.

$$1{,}250\text{mL} = 100\%$$

(You are converting the volume to a percentage.) Thus:

$$1\% = \frac{1{,}250}{100}$$

$$60\% = \frac{1{,}250}{100} \times 60 = 750\text{mL}$$

ANSWER: 60% of 1.25 litres is 750mL.

Alternatively, you can use the formula:

$$\text{amount} = \frac{\text{base}}{100} \times \text{per cent}$$

Re-writing this as:

$$\frac{\text{what you've got}}{100} \times \text{percentage required}$$

where 'what you've got' = 1,250mL and the percentage required = 60%.

Substitute the numbers into the formula:

$$\frac{1,250}{100} \times 60 = 750\text{mL}$$

ANSWER: 60% of 1.25 litres is 750mL

Now consider the following example:

WORKED EXAMPLE

What percentage is 125mg of 500mg?

Everything is already in the same units, milligrams (mg), so there is no need for any conversions. As always, the quantity you want to find the percentage of is equal to 100%, i.e.

500mg = 100% (You are converting the amount to a percentage.) Thus:

$$1\text{ mg} = \frac{100}{500}\%$$

$$125\text{ mg} = \frac{100}{500} \times 125 = 25\%$$

ANSWER: 125mg is 25% of 500mg.

Alternatively, you can use the formula:

$$\text{per cent} = \frac{100}{\text{base}} \times \text{amount} \quad \textit{or} \quad \frac{\text{amount}}{\text{base}} \times 100$$

where amount = 125mg and base = 500mg.

Substitute the numbers into the formula:

$$\frac{100}{500} \times 125 = 25\%$$

KEY POINTS

Per Cent

- Per cent means 'parts of a hundred' or a 'proportion of a hundred'.
- The symbol for per cent is %, so 30% means 30 parts or units of a hundred.
- Per centages are often used to give a quick indication of a specific quantity and are very useful when making comparisons.

Percentages and Fractions

- To convert a **fraction** to a **percentage**, **multiply** by 100.
- To convert a **percentage** to a **fraction**, **divide** by 100.

Percentages and Decimals

- To convert a **decimal** to a **percentage**, **multiply** by 100 – move the decimal point **two** places to the **right**.
- To convert a **percentage** to a **decimal**, **divide** by 100 – move the decimal point **two** places to the **left**.

Thus:

$$25\% = \frac{25}{100} = \frac{1}{4} = 0.25 \text{ (a quarter)}$$

$$33\% = \frac{33}{100} = \frac{1}{3} = 0.33 \text{ (approx. a third)}$$

$$50\% = \frac{50}{100} = \frac{1}{2} = 0.5 \text{ (a half)}$$

$$66\% = \frac{66}{100} = \frac{2}{3} = 0.66 \text{ (approx. two-thirds)}$$

$$75\% = \frac{75}{100} = \frac{3}{4} = 0.75 \text{ (three-quarters)}$$

To find the percentage of a number, always **divide** by 100:

$$\frac{\text{base}}{100} \times \text{per cent}$$

To find what percentage one number is of another, always **multiply** by 100:

$$\frac{\text{amount}}{\text{base}} \times 100$$

UNITS AND EQUIVALENCES

OBJECTIVES

At the end of this chapter, you should be familiar with the following:

• SI units

• Prefixes used in clinical medicine

• Equivalences

• Equivalences of weight

• Equivalences of volume

• Equivalences of amount of substance

• Conversion from one unit to another

• Writing units

INTRODUCTION

Many different units are used in medicine, for example:

• drug strengths, e.g. digoxin injection 500 micrograms in 1mL;

• dosages, e.g. dobutamine 3mcg/kg/min;

• patient electrolyte levels, e.g. sodium 137mmol/L.

It is important to have a basic knowledge of the units used in medicine and how they are derived. It is particularly important to have an understanding of the units in which drugs can be prescribed, and how to convert from one unit to another – this last part is very important as it forms the basis of all drug calculations.

SI UNITS

SI stands for *Système Internationale* (*d'Unités*), and is another name for the metric system of measurement. The aim of metrication is to make calculations easier than with the imperial system (ounces, pounds, stones, inches, pints, etc.). SI units are generally accepted in the United Kingdom and many other countries for use in medical practice and pharmacy. They were introduced in the NHS in 1975. SI base units The main units are those used to measure weight, volume and amount of substance:

• Weight: kilogram (kg)

• Volume: litre (l or L)

• Amount of substance: mole (mol)

SI prefixes

When an SI unit is inconveniently large or small, prefixes are used to denote multiples or sub-multiples. It is preferable to use multiples of a thousand. For example:

• gram

• milligram (one-thousandth, 1/1,000 of a gram)

• microgram (one-millionth, 1/1,000,000 of a gram)

• nanogram (one-thousand-millionth, 1/1,000,000,000 of a gram).

For example, one-millionth of a gram could be written as 0.000001g or as 1 microgram. The second version is easier to read than the first and easier to work with once you understand how to use units and prefixes. It is also

less likely to lead to errors, especially when administering drug doses.

PREFIXES USED IN CLINICAL MEDICINE

PREFIX	SYMBOL	DIVISION/MULTIPLE	FACTOR
Mega	M	× 1,000,000	10^{6}
Kilo	k	× 1,000	10^{3}
Deci	d	÷ 10	10^{-1}
Centi	c	÷ 100	10^{-2}
Milli	m	÷ 1,000	10^{-3}
Micro	mc (or μ)	÷ 1,000,000	10^{-6}
Nano	n	÷ 1,000,000,000	10^{-9}

The main prefixes you will come across on the ward are mega-, milli-, micro- and nano-. Thus, in practice, drug strengths and dosages can be expressed in various ways:

• Benzylpenicillin quantities are sometimes expressed in terms of megaunits (1 mega-unit means 1 million units of activity). Each vial contains benzylpenicillin 600mg, which equals 1 mega-unit.

• Small volumes of liquids are often expressed in millilitres (mL) and are used to describe small dosages, e.g. lactulose, 10mL to be given three times a day.

• Drug strengths are usually expressed in milligrams (mg), e.g. furosemide (frusemide) 40mg tablets.

• When the amount of drug in a tablet or other formulation is very small, strengths are expressed in micrograms or even nanograms, e.g. digoxin 125 microgram tablets; alfacalcidol 250 nanogram capsules.

EQUIVALENCES

The SI base units are too large for everyday clinical use, so they are subdivided into multiples of 1,000.

Equivalences of weight

Equivalences of weight

UNIT	SYMBOL	EQUIVALENT	SYMBOL
1 kilogram	kg	1,000 grams	g
1 gram	g	1,000 milligrams	mg
1 milligram	mg	1,000 micrograms	mcg
1 microgram	mcg	1,000 nanograms	ng



Equivalences of volume

UNIT	SYMBOL	EQUIVALENT	SYMBOL
1 litre	L or l*	1,000 millilitres	mL or ml

* *See section on 'Guide to writing units' below.*

Equivalences of amount of substance

UNIT	SYMBOL	EQUIVALENT	SYMBOL
1 mole	mol	1,000 millimoles	mmol
1 millimole	mmol	1,000 micromoles	mcmol

Examples of eqivalent amounts include:

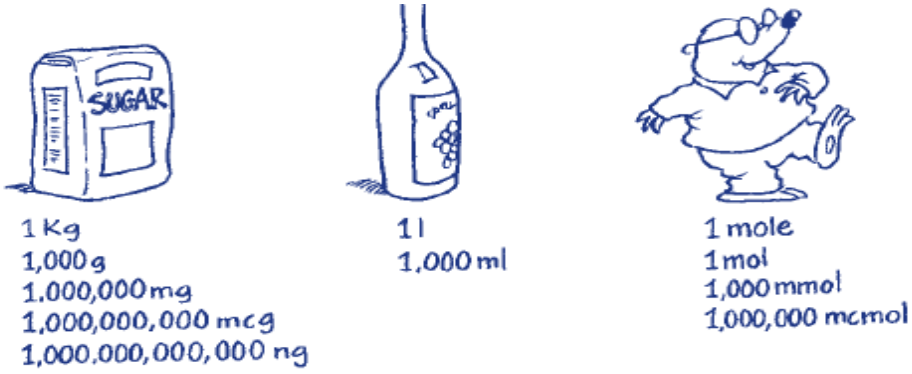
0.5 kg = 500g

0.25 g = 250mg

0.2mg = 200mcg

0.5L = 500mL

0.25mol = 250mmol



CONVERSION FROM ONE UNIT TO ANOTHER

In drug calculations, it is best to work in whole numbers, i.e. 125 micrograms rather than 0.125 mg, as fewer mistakes are then made.

Avoid using decimals, as the decimal point can be written in the wrong place during calculations. A decimal point in the wrong place can mean 10-fold or even 100-fold errors.

It is always best to work with the smaller unit in order to avoid decimals and decimal points, so you need to be able to convert easily from one unit to another. In general:

• To convert from a **larger** unit to a **smaller** unit, **multiply** by multiples of 1,000.

• To convert from a **smaller** unit to a **larger** unit, **divide** by multiples of 1,000.

For each multiplication or division by 1,000, the decimal point moves three places, either to the right or left depending upon whether you are converting from a larger unit to a smaller unit or vice versa. There are two methods for converting units: moving the decimal point or by using boxes which is an easy way to multiply or divide by a thousand (see the worked examples below). When you have to convert from a very large unit to a much smaller unit (or vice versa), you may find it easier to do the conversion in several steps.

For example, to convert 0.005 kg to milligrams, first convert to grams:

$$0.005 \text{ kg} = (0.005 \times 1{,}000)\text{g} = 5\text{g}$$

Next, convert grams to milligrams:

$$5\text{g} = (5 \times 1{,}000)\text{mg} = 5{,}000\text{mg}$$

WORKED EXAMPLE

Convert 0.5 g to milligrams.

METHOD ONE: MOVING THE DECIMAL POINT

You are going from a larger unit to a smaller unit, so you have to *multiply* by 1,000, i.e.

$$0.5 \times 1{,}000 = 500 \text{ milligrams}$$

The **decimal point** moves **three** places to the **right**:





GUIDE TO WRITING UNITS

The *British National Formulary* makes the following recommendations:

- The unnecessary use of decimal points should be avoided, e.g. 3 mg, **not** 3.0 mg.
- Quantities of 1 gram or more should be expressed as 1.5 g, etc.
- Quantities less than 1 gram should be written in milligrams, e.g. 500 mg, **not** 0.5 g.
- Quantities less than 1 mg should be written in micrograms, e.g. 100 micrograms, **not** 0.1 mg.
- When decimals are unavoidable, a zero should be written in front of the decimal point where there is no other figure, e.g. 0.5 mL **not** .5 mL. However, the use of a decimal point is acceptable to express a range, e.g. 0.5–1 g.
- Micrograms and nanograms should **not** be abbreviated. Similarly, 'units' should **not** be abbreviated.
- A capital L is used for litre, to avoid confusion (a small letter l could be mistaken for a figure 1 (one), especially when typed or printed).
- Cubic centimetre, or cm^3, is **not** used in medicine or pharmacy; use millilitre (mL or ml) instead.

PROBLEMS

Question 1 Convert 0.0125 kilograms to grams.
Question 2 Convert 250 nanograms to micrograms.
Question 3 Convert 3.2 litres to millilitres.
Question 4 Convert 0.0273 moles to millimoles.
Question 5 Convert 3,750 grams to kilograms.
Question 6 Convert 0.05 grams to micrograms.
Question 7 Convert 25,000 milligrams to kilograms.
Question 8 Convert 4.5 × 10–6 grams to nanograms.
Question 9 You have an ampoule of digoxin 0.25mg/mL.
Calculate the amount of digoxin inmicrograms in a 2mL ampoule.
Question 10 You have an ampoule of fentanyl 0.05mg/mL.
Calculate the amount of fentanyl in micrograms in a 2mL

ampoule.

The following two case reports illustrate how bad writing can lead to problems.

CONFUSING MICROGRAMS AND MILLIGRAMS

Case report

On admission to hospital, a patient taking thyroxine replacement therapy presented her general practitioner's referral letter which stated that her maintenance dose was 0.025 mg once daily. The clerking house officer incorrectly converted this dose and prescribed 250 micrograms rather than the 25 micrograms required. A dose was administered before the error was detected by the ward pharmacist the next morning.

Taken from: *Pharmacy in Practice 1994; 4: 124.*

This example highlights several errors:

- The wrong units were originally used – milligrams instead of micrograms.
- A number containing a decimal points was used.
- Conversion from one unit to another was carried out incorrectly.

THIS UNIT ABBREVIATION IS DANGEROUS

Case report

Patient received 50 units of insulin instead of the prescribed stat dose of 5 units. A junior doctor requiring a patient to be given a stat dose of 5 units Actrapid insulin wrote the prescription appropriately but chose to incorporate the abbreviation ⊙ for 'units', which is occasionally seen used on written requests for units of blood. Thus the prescription read: 'Actrapid insulin 5⊙ stat'.

The administering nurse misread the abbreviation and interpreted the prescription as 50 units of insulin. This was administered to the patient, who of course became profoundly hypoglycaemic and required urgent medical intervention.

Comment

The use of the symbol ⊙ to indicate units of blood is an old-fashioned practice which is now in decline. This case serves to illustrate the catastrophic effect that the inappropriate use of this abbreviation can have – it led to misinterpretation by the nursing staff and resulted in harm to the patient.

Taken from: *Pharmacy in Practice 1995; 5: 131.*

DRUG STRENGTHS OR CONCENTRATIONS

OBJECTIVES

At the end of this chapter, you should be familiar with the following:

• Percentage concentration

• Calculating the total amount of drug in a solution

• mg/ml concentration

• Converting percentage to a mg/ml concentration

• '1 in ...' concentrations or ratio strengths

• Parts per million (ppm)

• Drug concentrations expressed in units: heparin and insulin

INTRODUCTION

There are various ways of expressing how much actual drug is present in a medicine. These medicines are usually liquids that are for oral or parenteral administration, but also include those for topical use.

PERCENTAGE CONCENTRATION

Following on from the last chapter, one method of describing concentration is to use the percentage as a unit. Percentage concentration can be defined as the amount of drug in 100 parts of the product. The most common method you will come across is the percentage concentration w/v (weight in volume). This is used when a solid is dissolved in a liquid and means the number of grams dissolved in 100mL:

% w/v = number of grams in 100mL

(Thus 5% w/v means 5g in 100mL.)

Another type of concentration you might come across is the percentage concentration w/w (weight in weight). This is used when asolid is mixed with another solid, e.g. in creams and ointments, and means the number of grams in 100g:

% w/w = number of grams in 100 g

(Thus 5% w/w means 5g in 100g.)

Another is the percentage concentration v/v (volume in volume). This is used when a liquid is mixed or diluted with another liquid, and means the number of millilitres (mL) in 100mL:

% v/v = number of mL in 100mL

(Thus 5% v/v means 5mL in 100mL.)

The most common percentage concentration you will encounter is the percentage w/v or 'weight in volume' and therefore this will be the one considered here.

In our earlier example of 5% w/v, there are 5g in 100mL irrespective of the size of the container. For example, glucose 5% infusion means that there are 5g of glucose dissolved in each 100mL of fluid and this will remain the same if it is a 500mL bag or a 1 litre bag.

To find the total amount of drug present in a bottle or infusion bag, you must take into account the size or volume of the bottle or infusion bag.

WORKED EXAMPLE

TO CALCULATE THE TOTAL AMOUNT OF DRUG IN A SOLUTION

How much sodium bicarbonate is there in a 200mL infusion sodium bicarbonate 8.4% w/v?

STEP ONE

Convert the percentage to the number of grams in 100mL, i.e. 8.4% w/v = 8.4 g in 100mL

(You are converting the percentage to a specific quantity.)

STEP TWO

Calculate how many grams there are in 1mL, i.e. divide by 100:

$$\frac{8.4\,g}{100} \text{ in } 1\,mL \text{ (using the 'ONE unit' rule)}$$

STEP THREE

However, you have a 200mL infusion. So to find out the total amount present, multiply how much is in 1mL by the volume you've got (200mL):

$$\frac{8.4}{100} \times 200 = 16.8\,g \text{ in } 200\,mL$$

ANSWER: There are 16.8 g of sodium bicarbonate in 200mL of sodium bicarbonate 8.4% w/v infusion. A simple formula can be devised based upon the formula seen earlier:

$$\text{amount} = \frac{\text{base}}{100} \times \text{per cent}$$

This can be re-written as:

$$\text{Total amount (g)} = \frac{8.4}{100} \times 200 = 16.8\,g$$

ANSWER: There are 16.8 g of sodium bicarbonate in 200mL of sodium bicarbonate 8.4% w/v infusion.

PROBLEMS

Calculate how many grams there are in the following:

Question 1 How many grams of sodium chloride are there in a litre infusion of sodium chloride 0.9%?

Question 2 How many grams of potassium, sodium and glucose are there in a litre infusion of potassium 0.3%, sodium chloride 0.18% and glucose 4%?

Question 3 How many grams of sodium chloride are there in a 500mL infusion of sodium chloride 0.45%?

Question 4 You need to give calcium gluconate 2 g as a slow intravenous injection. You have 10mL ampoules of calcium gluconate 10%. How much do you need to draw up?

mg/mL CONCENTRATIONS

Another way of expressing the amount or concentration of drug in a solution, usually for oral or parenteral administration, is in mg/mL, i.e. number of milligrams of drug per mL of liquid. This is the most common way of expressing the amount of drug in a solution.

For oral liquids, it is usually expressed as the number of milligrams in a standard 5mL spoonful, e.g. amoxicillin (amoxycillin) 250mg in 5mL.For oral doses that are less than 5mL an oral syringe would be used (see the section 'Administration of medicines' in Chapter 9 'Action and administration of medicines', page xx).

For injections, it is usually expressed as the number of milligrams per volume of the ampoule (1mL, 2mL, 5mL, 10mL and 20mL), e.g. gentamicin 80mg in 2mL. However, injections may still be expressed as the number of milligrams per mL, e.g. furosemide (frusemide) 10mg/mL, 2mL – particularly in old reference sources.

Note: Strengths can also be expressed in mcg/mL, e.g. hyoscine injection 600mcg/1mL. Only mg/mL will be considered here, but the principles learnt here can be applied to other concentrations or strengths, e.g. mcg/mL. Sometimes it may be useful to convert percentage concentrations to mg/mL concentrations. For example:

lidocaine 0.2% (lignocaine) = 0.2 g per 100mL

	= 200mg per 100mL
	= 2mg per mL (2mg/mL)
sodium chloride 0.9%	= 0.9 g per 100mL
	= 900mg per 100mL
	= 9mg per mL (9mg/mL)
glucose 5%	= 5g per 100mL
	= 5,000mg per 100mL
	= 50mg per mL (50mg/mL)

This will give the strength of the solution irrespective of the size of the bottle, infusion bag, etc.

If you know the percentage concentration, an easy way of finding the strength in mg/mL is by simply multiplying the percentage by 10. This can be explained using lidocaine (lignocaine) 0.2% as an example: You have lidocaine (lignocaine) 0.2% – this is equal to 0.2 g in 100mL. Divide by 100 to find out how much is in 1mL. Thus:

$$\frac{0.2}{100}\ \text{g/mL}$$

Multiply by 1,000 to convert grams to milligrams. Thus:

$$\frac{0.2}{100} \times 1{,}000\ \text{mg/mL}$$

.Simplify the above calculation to give:

$$0.2 \times 10 = 2\text{mg/mL}$$

Therefore you simply multiply the percentage by 10.

With the other examples above:

sodium chloride 0.9% $0.9 \times 10 = 9\text{mg/mL}$

glucose 5% $5 \times 10 = 50\text{mg/mL}$

Conversely, to convert a mg/mL concentration to a percentage, you simply divide by 10.

Once again, if we use our original lidocaine (lignocaine) as an example: You have lidocaine (lignocaine) 2mg/mL. Percentage means ‘per 100mL’, so multiply by 100, i.e.

$$2\text{mg/mL} \times 100 = 200\text{mg/100mL}\ (2 \times 100)$$

Remember, percentage (w/v) means ‘the number of grams per 100 mL’, so you will have to convert milligrams to grams by dividing by 1,000, i.e.

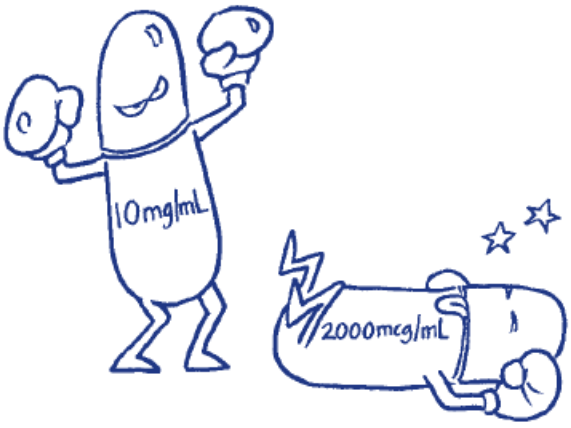
$$\frac{2\times100}{1{,}000}=\frac{2}{10}=0.2\%$$

Therefore to change a strength in mg/mL to a percentage strength you simply divide the mg/mL concentration by 10.

With our other examples:

sodium chloride 9mg/mL 9 ÷ 10 = 0.9%

glucose 50mg/mL 50 ÷ 10 = 5%



PROBLEMS

Calculate the strengths (mg/mL) for the following:

.Question 5 Sodium chloride infusion 0.45%

Question 6 Metronidazole infusion 0.5%

Question 7 Potassium chloride 0.2%, sodium chloride 0.18% and glucose 4%

Convert the following mg/mL strengths to percentage strengths:

Question 8 Bupivacaine 2.5mg/mL

Question 9 Glucose 500mg/mL

Question 10 Isosorbide dinitrate 500 micrograms/mL

'1 IN ...' CONCENTRATIONS OR RATIO STRENGTHS

This style of expressing concentration is used only occasionally, and is written as '1 in ...', e.g. 1 in 10,000, and is sometimes known as a ratio strength. It usually refers to a solid dissolved in a liquid and, by agreed convention, the weight is expressed in grams and the volume in milliliters.

So, it means **one** gram in however many mL. For example:

'1 in 1,000' means 1 g in 1,000mL

'1 in 10,000' means 1 g in 10,000mL

Therefore it can be seen that 1 in 10,000 is weaker than 1 in 1,000. So, the higher the 'number', the weaker the solution.

The drug most commonly expressed this way is adrenaline/epinephrine:

Adrenaline/epinephrine 1 in 1,000 which is equal to 1*mg* in 1mL

Adrenaline/epinephrine 1 in 10,000 which is equal to 1*mg* in 10mL

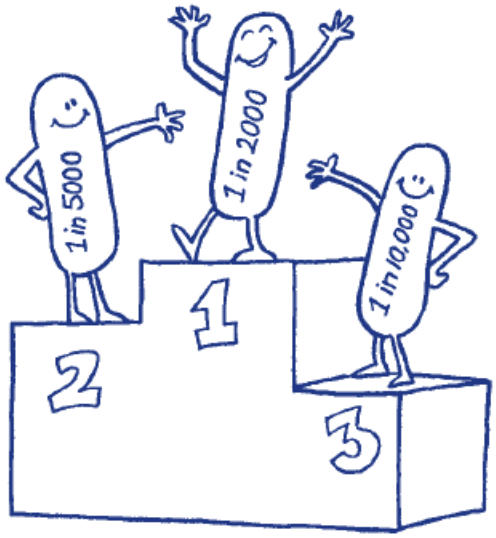
An easy way to remember the above is to cancel out the three zeros that appear after the comma, i.e.

Adrenaline/epinephrine 1 in 1,000 –

cancel out the three zeros after the comma: 1,0/ 0/ 0/, to give:

1 in 1 which can be written as: 1mg in 1mL

Similarly, for adrenaline/epinephrine 1 in 10,000 – cancel out the three zeros after the comma: 10,000 to give: 1 in 10, which can be written as 1mg in 10mL.



PARTS PER MILLION (ppm)

This is a way of expressing very dilute concentrations. Just as per cent means parts of a hundred, so parts per million or ppm means parts of a million. It usually refers to a solid dissolved in a liquid but, as with percentage concentrations, it can also be used for two solids or two liquids mixed together.

Once again, by agreed convention:

1 ppm means 1 g in 1,000,000mL or 1mg in 1 litre (1,000mL)

In terms of percentage, 1 ppm equals 0.0001%.

Other equivalents include:

One part per million is one second in 12 days of your life!

One part per million is one penny out of £10,000!

For example, solutions produced by disinfectant chlorine-releasing agents (e.g. Haz-Tabs®) are measured in terms of parts per million, such as 1,000 ppm available chlorine.

PROBLEMS

Question 11 Adrenaline/epinephrine is sometimes combined with lidocaine (lignocaine) when used as a local anaesthetic, usually at a 1 in 200,000 strength. How much adrenaline/epinephrine is there in a 20mL vial?

Question 12 It is recommended that children should have fluoride supplements for their teeth if the fluoride content of drinking water is 0.7ppm or less. Express this concentration in micrograms per litre.

Calculating Insulin Dose

At the end of this section , you should be familiar with the following:

- An example of how to calculate a carbohydrate coverage dose
- An example of how to calculate a high blood sugar correction dose
- An example of how to calculate a mealtime insulin dose
- Formulas that are commonly used by providers to create insulin doserecommendations

First, some basic things to know about insulin:

- Approximately 40-50% of the total daily insulin dose is to replace insulin overnight, when you are fasting and between meals. This is called background or basal insulin replacement. The basal or background insulin dose usually is constant from day to day.
- The other 50-60% of the total daily insulin dose is for carbohydrate coverage (food) and high blood sugar correction. This is called the bolus insulin replacement.

Bolus – Carbohydrate coverage

The bolus dose for food coverage is prescribed as an insulin to carbohydrate ratio.The insulin to carbohydrate ratio represents how many grams of carbohydrate are covered or disposed of by 1 unit of insulin.

Generally, one unit of rapid-acting insulin will dispose of 12-15 grams of carbohydrate. This range can vary from 4-30 grams or more of carbohydrate depending on an individual's sensitivity to insulin. Insulin sensitivity can vary according to the time of day, from person to person, and is affected by physical activity and stress.

Bolus – High blood sugar correction

(also known as insulin sensitivity factor)

The bolus dose for high blood sugar correction is defined as how much one unit of rapid-acting insulin will drop the blood sugar**.**

Generally, to correct a high blood sugar, one unit of insulin is needed to drop the blood glucose by 50 mg/dl. This drop in blood sugar can range from 15-100 mg/dl or more, depending on individual insulin sensitivities, and other circumstances.

Examples:

Read some examples and therapeutic principles on how to calculate the carbohydrate coverage dose, high blood sugar correction dose and the total mealtime insulin dose.

Example #1: Carbohydrate coverage at a meal

First, you have to calculate the carbohydrate coverage insulin dose using this formula:

CHO insulin dose =

Total grams of CHO in the meal

÷ grams of CHO disposed by 1 unit of insulin

(the grams of CHO disposed of by 1 unit of insulin is the bottom number or denominator of the Insulin:CHO ratio).

For Example #1, assume:

- You are going to eat 60 grams of carbohydrate for lunch
- Your Insulin: CHO ratio is 1:10

To get the CHO insulin dose, plug the numbers into the formula:

CHO insulin dose =

Total grams of CHO in the meal (60 g)

÷ grams of CHO disposed by 1 unit of insulin (10) = 6 units

You will need 6 units of rapid acting insulin to cover the carbohydrate.

Example #2: High blood sugar correction dose

Next, you have to calculate the high blood sugar correction dose.

High blood sugar correction dose =

Difference between actual blood sugar and target blood sugar*

÷ correction factor.

*Actual blood sugar minus target blood sugar

For Example #2, assume:

- 1 unit will drop your blood sugar 50 points (mg/dl) and the high blood sugar correction factor is 50.
- Pre-meal blood sugar target is 120 mg/dl.
- Your actual blood sugar before lunch is 220 mg/dl.

Now, calculate the difference between your actual blood sugar and target blood sugar:

220 minus 120 mg/dl = 100 mg/dl

To get the high blood sugar correction insulin dose, plug the numbers into this formula:

Correction dose =

Difference between actual and target blood glucose (100mg/dl)

÷ correction factor (50) = 2 units of rapid acting insulin

So, you will need an additional 2 units of rapid acting insulin to "correct" the blood sugar down to a target of 120 mg/dl.

Example #3: Total mealtime dose

Finally, to get the total mealtime insulin dose, add the CHO insulin dose together with the high blood sugar correction insulin dose:

CHO Insulin Dose

+ High Blood Sugar Correction Dose

= Total Meal Insulin Dose

For Example #3, assume:

- The carbohydrate coverage dose is 6 units of rapid acting insulin.
- The high blood sugar correction dose is 2 units of rapid acting insulin.

Now, add the two doses together to calculate your total meal dose.

Carbohydrate coverage dose (6 units)

+ high sugar correction dose (2 units)

= 8 units total meal dose!

The total lunch insulin dose is 8 units of rapid acting insulin.

Example #4: Formulas commonly used to create insulin dose recommendations

This example illustrates a method for calculating of your background/basal and bolus doses and estimated daily insulin dose when you need full insulin replacement. Bear in mind, this may be too much insulin if you are newly diagnosed or still making a lot of insulin on your own. And it may be too little if you are very resistant to the action of insulin. Talk to your provider about the best insulin dose for you as this is a general formula and may not meet your individual needs.

The initial calculation of the basal/background and bolus doses requires estimating your total daily insulin dose:

Total Daily Insulin Requirement

The general calculation for the body's daily insulin requirement is:

Total Daily Insulin Requirement(in units of insulin)

= Weight in Pounds ÷ 4

Alternatively, if you measure your body weight in kilograms:

Total Daily Insulin Requirement (in units of insulin)

= 0.55 X Total Weight in Kilograms

Example 1:

If you are measuring your body weight in pounds:

- Assume you weigh 160 lbs.

In this example:

TOTAL DAILY INSULIN DOSE = 160 lb ÷ 4 = 40 units of insulin/day

Example 2:

If you are measuring your body weight in kilograms:

- Assume your weight is 70Kg

In this example:

TOTAL DAILY INSULIN DOSE

= 0.55 x 70 Kg = 38.5 units of insulin/day

If your body is very resistant to insulin, you may require a higher dose. If your body is sensitive to insulin, you may require a lower insulin dose.

Basal/Background and Bolus Insulin Doses

Next, you need to establish the basal/background dose, carbohydrate coverage dose (insulin to carbohydrate ratio) and high blood sugar correction dose (correction factor).

Basal/background insulin dose:

Basal/background Insulin Dose

= 40-50% of Total Daily Insulin Dose

Example

1. Assume you weigh 160 pounds
2. Your total daily insulin dose (TDI) = 160 lbs ÷ 4 = 40 units.

In this example:

Basal/background insulin dose

= 50% of TDI (40 units) = 20 units

of either long acting insulin,(such as glargine or detemir) or rapid acting insulin if you are using an insulin pump (continuous subcutaneous insulin infusion device).

The carbohydate coverage ratio:

500 ÷ Total Daily Insulin Dose

= 1 unit insulin covers so many grams of carbohydrate

This can be calculated using the Rule of "500": Carbohydrate Bolus Calculation

Example:

1. Assume your total daily insulin dose (TDI)
   = 160 lbs ÷ 4 = 40 units

In this example:

Carbohydrate coverage ratio

= 500 ÷ TDI (40 units)

= 1unit insulin/ 12 g CHO

This example above assumes that you have a constant response to insulin throughout the day. In reality, individual insulin sensitivity varies. Someone who is resistant in the morning, but sensitive at mid-day, will need to adjust the insulin-to-carbohydrate ratio at different meal times. In such a case, the background insulin dose would still be approximately 20 units; however, the breakfast insulin-to-carbohydrate ratio might be breakfast 1:8 grams, lunch 1:15 grams and dinner 1:12 grams.

the insulin to carbohydrate ratio may vary during the day.

The high blood sugar correction factor:

Correction Factor = 1800 ÷Total Daily Insulin Dose = 1 unit of insulin will reduce the blood sugar so many mg/dl

This can be calculated using the Rule of "1800".

Example:

1. Assume your total daily insulin dose(TDI) = 160 lbs ÷ 4 = 40 units

In this example:

Correction Factor

= 1800 ÷ TDI(40 units)

= 1 unit insulin will drop reduce the blood sugar level by 45 mg/dl

While the calculation is 1 unit will drop the blood sugar 45 mg/dl, to make it easier most people will round up or round down the number so the suggested correction factor may be 1 unit of rapid acting insulin will drop the blood sugar 40-50 mg/dl.

KEY POINTS

Percentage Concentration

% w/v = number of grams in 100 mL

(A solid is dissolved in a liquid, thus 5% w/v means 5 g in 100 mL.)

% w/w = number of grams in 100 g

(A solid mixed with another solid, thus 5% w/w means 5 g in 100 g.)

%v/v = number of mL in 100 mL

(A liquid is mixed or diluted with another liquid, thus 5% v/v means 5 mL in 100 mL.)

- Most common percentage strength encountered is % w/v.
- There will always be the same amount of drug present in 100 mL irrespective of the total volume. Thus in a 5% w/v solution, there is 5 g dissolved in each 100 mL of fluid and this will remain the same if it is a 500 mL bag or a 1 litre bag.
- To find the total amount of drug present, the total volume must be taken into account – in 500 mL of a 5% w/v solution there is a total of 25 g present.

mg/mL Concentrations

- Defined as the number of milligrams of drug per millilitre of liquid.
- Oral liquids – usually expressed as the number of mg in a standard 5 mL spoonful, e.g. erythromycin 250 mg in 5 mL.
- Injections are usually expressed as the number of mg per volume of the ampoule (1 mL, 2 mL, 5 mL, 10 mL or 20 mL), e.g. gentamicin 80 mg in 2 mL.

Strengths can also be expressed in mcg/mL.

Converting percentage concentrations to mg/mL concentrations

- Multiply the percentage by 10, e.g. lidocaine (lignocaine) 0.2% = 2 mg/mL.

Converting mg/mL concentrations to percentage concentrations

- Divide the mg/mL strength by 10, e.g. lidocaine (lignocaine) 2 mg/mL = 2%.

'1 in ...' Concentrations or Ratio Strengths

- Defined as: **one** gram in however many millilitres.

For example:

1 in 1,000 means 1 g in 1,000 mL
1 in 10,000 means 1 g in 10,000 mL

Parts per Million (ppm)

- Similar to ratio strengths, but used to describe very dilute concentrations.
- Most common ppm concentration encountered is that of a solid dissolved in a liquid, but can also apply to two solids or liquids mixed together.
- Defined as: **one** gram in 1,000,000 mL *or* **one** milligram in 1 litre.

Units – Heparin and Insulin

- The purity of drugs such as insulin and heparin from animal or biosynthetic sources varies.
- Therefore these drugs are expressed in terms of **units** as a standard measurement rather than weight.

DOSAGE CALCULATIONS

OBJECTIVES

At the end of this chapter, you should be familiar with the following:

• Calculating the number of tablets or capsules required

• Dosages based on patient parameters

• Ways of expressing doses

• Calculating dosages

• Displacement values or volumes

What is displacement?

Is displacement important in medicine?

INTRODUCTION

Dosage calculations are the basic everyday type of calculations you will be doing on the ward. They include calculating number of tablets or capsules required, divided doses, simple drug dosages and dosages based on patient parameters, e.g. weight and body surface area.

It is important that you are able to do these calculations confidently, as mistakes may result in the patient receiving the wrong dose which may lead to serious consequences for the patient. After completing this chapter, should you not only be able to do the calculations, but also be able to decide whether your answer is reasonable or not.

CALCULATING THE NUMBER OF TABLETS OR CAPSULES REQUIRED

On the drug round, you will usually have available the strength of the tablets or capsules for the dose prescribed on a patient's drug chart, e.g. dose prescribed is furosemide (frusemide) 40 mg; tablets available are furosemide (frusemide) 40 mg. However, there may be instances when the not match the dose prescribed. Then you will have to calculate how many tablets or capsules to give the patiet.

WORKED EXAMPLES

A patient is prescribed 75mcg of levothyroxine sodium (thyroxine sodium) but the strength of the tablets available is 25 mcg. How many tablets are required?

This is a very simple calculation. The answer involves finding how many 25s there are in 75 or in other words 75 divided by 25:

$$\frac{75}{25} = \frac{3}{1} = 3 \text{ tablets}$$

In most cases, it is a simple sum you can do in your head, but even so, it is a drug calculation so care must always be taken. A patient is prescribed 2 g of flucloxacillin to be given orally but it is available in 500mg capsules. How many capsules should you give? Once again it is a simple calculation but it is slightly more complicated than our earlier example as the dose prescribed and the available medication are in different units.

The first step is to convert everything to the same units. We could either convert the 500 mg into grams, or we could convert the 2 g into milligrams.

In this case it is preferable to convert the grams to milligrams as this avoids decimal points. Remember it is best not to work with decimal points – a decimal point in the wrong place can mean a 10-fold or even a 100-fold error.

To convert grams to milligrams, multiply by 1,000:

$$2 \text{ g} = (2 \times 1{,}000)\text{mg} = 2{,}000\text{mg}$$

The calculation is now similar to our earlier example. The answer involves finding how many 500s are in 2,000 or in other words 2,000 divided by 500:

$$\frac{2{,}000}{500} = \frac{4}{1} = 4 \text{ capsules}$$

Once again, it is a simple sum you can do in your head, but it is a drug

calculation, so care must always be taken.

A formula can be derived:

A formula can be derived:

$$\text{number required} = \frac{\text{amount prescribed}}{\text{amount in each tablet or capsule}}$$

PROBLEMS

Question 1 500 mg is prescribed and the tablets are 250 mg each. How many tablets will you give?

Question 2 Alfacalcidol 1 microgram is prescribed. If you only have 250 nanogram capsules, how many would you give?

DOSAGES BASED ON PATIENT PARAMETERS

Sometimes, the dose required is calculated on a body weight basis (mg/kg) or in terms of a patient's surface area (mg/m2). Using body surface area (BSA) estimates is more accurate than using body weight, since many physical phenomena are more closely related to BSA. This particularly applies to cytotoxics and other drugs that require an accurate individual dose. To find the BSA for a patient, you will need to know that patient's height and weight. Then the BSA can be calculated, using a formula or nomograms

$$m^2 = \sqrt{\frac{\text{height (cm)} \times \text{weight (kg)}}{3{,}600}}$$

For example, you want to know the body surface area of a child whose weight is 16.4 kg and height is 100 cm. Substitute the figures into the formula:

$$\text{BSA} = \sqrt{\frac{100 \times 16.4}{3{,}600}}$$
$$= \sqrt{\frac{1{,}640}{3{,}600}}$$
$$= \sqrt{0.456}$$
$$= 0.675\,m^2$$

WORKED EXAMPLES

WEIGHT

The dose required is 3 mg/kg and the patient weighs 68 kg.

This means that for every kilogram (kg) of a patient's weight, you will need 3mg of drug.

In this example, the patient weighs 68 kg. Therefore this patient will need 68 lots of 3mg of drug, i.e. you simply multiply the dose by the patient's weight:

$$3 \times 68 = 204\text{mg}$$

Thus the patient will need a total dose of 204mg.
This can be summarized as:

total dose required = dose per kg × patient′s weight

SURFACE AREA

Doses are calculated in the same way, substituting surface area for weight. For example: the dose required is 500mg/m2 and the patient's body surface area is 1.89m2.
For every square metre (m2) of a patient's surface area, you will need 500mg of drug.
In this example, the patient's body surface area is 1.89m2. Therefore this patient will need 1.89 lots of 500 mg of drug, i.e. you simply multiply the dose by the patient's body surface area (obtained from a formula or nomograms – see Appendix 1).

$$500 \times 1.89 = 945\text{mg}$$

Thus the patent will need a total dose of 945mg.

This can be summarized as:

total dose required = dose per m2 × body surface area

When using this method of calculation, the actual body weight should be used. However, in the case of obese children, the child may receive an artificially high dose. The reason for this is that fat tissue plays virtually no part in metabolism, and the dose must be estimated on lean or ideal body weight. As a rule of thumb, doses should be reduced by approximately 25% for obese children.

PROBLEMS

Work out the following dosages:

Question 3 Dose = 1.5 mg/kg, patient′s weight = 73 kg

Question 4 Dose = 60 mg/kg, patient′s weight = 12 kg

Question 5 Dose = 50 mg/m2, patient′s surface area = 1.94m2

Question 6 Dose = 120 mg/m2, patient′s surface area = 1.55m2

Question 7 Dose = 400 mcg/kg, patient′s weight = 54 kg

i) What is the total dose in micrograms?

ii) What is the total dose in mg?

Question 8 Dose = 5 mcg/kg/min, patient's weight = 65 kg

What is the dose in mcg/min? (You will meet this type of calculation with IV infusions

The following table will be needed to answer questions 9 and 10.

LMWH	STRENGTH	PREPARATIONS AVAILABLE
Enoxaparin (*Clexane*®)	100 mg/mL (10,000 units/mL)	0.2 mL syringe (2,000 units, 20 mg) 0.4 mL syringe (4,000 units, 40 mg) 0.6 mL syringe (6,000 units, 60 mg) 0.8 mL syringe (8,000 units, 80 mg) 1 mL syringe (10,000 units, 100 mg)
Tinzaparin (*Innohep*®)	20,000 units/mL	0.5 mL syringe (10,000 units) 0.7 mL syringe (14,000 units) 0.9 mL syringe (18,000 units) 2 mL vial (40,000 units)

Question 9 A patient has a deep vein thrombosis (DVT) and needs to be given tinzaparin at a dose of 175 units/kg. The patient weighs 68 kg. What dose do they need, which syringe do you use and volume you give?

Question 10 A patient has a DVT and has been prescribed enoxaparin 1.5 mg/kg. The patient weighs 59 kg. What dose do they need, which syringe do you use and volume you give?

Question 11 Using the Mosteller BSA formula (see Appendix 1):

$$\text{BSA } m^2 = \sqrt{\frac{\text{height (cm)} \times \text{weight (kg)}}{3{,}600}}$$

find out the body surface area for a child weighing 20 kg with a height of 108 cm.

Question 12 Using the Mosteller BSA formula (see Appendix 1):

$$\text{BSA } m^2 = \sqrt{\frac{\text{height (cm)} \times \text{weight (kg)}}{3{,}600}}$$

Find out the body surface area for a patient weighing 96 kg and with a height of 180 m.

WAYS OF EXPRESSING DOSES

A dose is the quantity or amount of a drug taken by, or administered to, a patient to achieve a therapeutic outcome. A dose can be described as a single dose, a daily dose, a daily divided dose, a weekly dose or a total dose, etc., as described in the examples below:

• Single dose: for example, pre-medication drugs – this is sometimes referred to as a 'stat' dose, meaning 'at once' from the Latin *statum*.

• Daily dose: for example, the BNF recommended dose of atorvastatin is 10mg once daily.
• Weekly dose: for example, methotrexate, when used in rheumatoid arthritis.
• Divided or total daily dose: for example, the *BNF for Children* recommends for cefradine (cephradine):

By mouth

Child 1 month–12 years

12.5–25 mg/kg twice daily (total daily dose may alternatively be given in three or four divided doses)

As stated above, the dose of a drug can be given as a total daily dose (TDD), which has to be given in divided doses (usually three or four times a day). This is particularly associated with paediatric doses. It is important that you can tell the difference between the TDD and individual doses. If not interpreted properly, then the patient is at risk of receiving the TDD as an individual dose, thus receiving three or four times the normal dose (with potentially disastrous results). Consider the above example for a 6-year-old (weight = 20.5 kg).
If giving a dose of cefradine (cephradine) 12.5 mg/kg, then the dose is 12.5 × 20.5 = 256mg (256 mg 'rounded up'). Consider the above example for a 6 year old weight = 20.5kg)

If giving a dose of cefradine (cephradine) 12.5 mg/kg, then the dose is 12.5 × 20.5 =

(256.25mg 'rounded down'). As this dose is to be given 'twice daily', the TDD will be

256 ×2 = 512 mg.

If using a 125 mg/5mL suspension, it would be appropriate to give this in four divided doses:
512 mg = 20.48 mL cefradine (cephradine) 125 mg/5 mL suspension
So the dose is: 20.48 /4 = 5.12 mL (5 ml) FOUR times a day.

CALCULATING DRUG DOSAGES

There are several ways of solving this type of calculation. It is best to learn one way and stick to it. The easiest way is by proportion: what you do to one side of an equation, do the same to the other side. The 'ONE unit' rule described in Chapter 1 'First principles' will be used here. Also, when what you've got and what you want are in different units, you need to convert everything to the same units. When converting to the same units, it is best to convert to whole numbers to avoid decimal points, as fewer mistakes are then made. If possible, it is a good idea to convert everything to the units of the answer.

WORKED EXAMPLE

You need to give a patient 125 micrograms of digoxin orally. You have digoxin elixir 50 micrograms/mL supplied with a dropper pipette. How much do you need to draw up?

Before we continue with the calculation, we will try to estimate our answer. This is only really possible with such simple calculations.

We have:	50 micrograms in 1 mL	
So:	100 micrograms in 2 mL	(by doubling)
It follows:	150 micrograms in 3 mL	(1 mL + 2 mL)
From this:	125 micrograms would be within the range 2–3 mL.	

STEP ONE

Write down what you have: 50 micrograms in 1 mL

STEP TWO

Calculate how much ONE unit is of what you have, i.e.:

$$50 \text{ micrograms in } 1 \text{ mL}$$

$$1 \text{ microgram} = \frac{1}{50} \text{mL}$$

This is the 'ONE unit' rule.

STEP THREE

You need to know how much digoxin to draw up for 125 micrograms, therefore multiply the amount from Step Two by 125:

$$125 \text{ micrograms} = \frac{1}{50} \times 125 = 2.5 \text{mL}$$

ANSWER: You will need to draw up 2.5 mL of digoxin.

This correlates with our estimation of our answer; 2.5 mL is within the range 2 to 3 mL. In fact, we could have easily worked out the answer exactly from our estimations.

From the above, a formula can be derived to calculate drug dosages:

$$\frac{\text{amount you want}}{\text{amount you've got}} \times \text{volume it's in}$$

This formula should be familiar as this is the one universally taught for calculating doses. Remember care must be taken when using any formula – ensure that numbers are entered and calculated correctly. From the above example:

$$\text{amount you want} = 125 \text{ micrograms}$$
$$\text{amount you've got} = 50 \text{ micrograms}$$
$$\text{volume it's in} = 1 \text{ mL}$$

Substitute the numbers in the formula:

$$\frac{125}{50} \times 1 = 2.5 \text{mL}$$

ANSWER: You will need to draw up 2.5 mL of digoxin.

You can apply this method to whatever type of calculation you want.

The following case report illustrates the importance of ensuring that your calculations are right.

A PROBLEM WITH A DECIMAL POINT

Case report

A female baby, born seven weeks prematurely, died at 28 hours old when a junior doctor miscalculated a dose of intravenous morphine, resulting in the administration of a 100-times overdose. The doctor is reported to have worked out the dose on a piece of paper and then checked it on a calculator but the decimal point was inserted in the wrong place and 15 instead of 0.15 milligrams was prescribed. The dose was then prepared and handed to the senior registrar who administered it without double-checking the calculation and, despite treatment with naloxone, the baby died 55 minutes later.

Taken from: *Pharmacy in Practice* 1997; **7**: 368–9.

The following two cases illustrate the importance of checking numbers before administration.

BE ALERT TO HIGH NUMBERS OF DOSE UNITS

Case report

Case One

A male patient was prescribed a stat dose of 2g amiodarone for conversion of atrial fibrillation. Although it is still not known whether this dose was chosen deliberately or prescribed in error, there is evidence to support the use of a 2g oral regimen. What concerned the reporting hospital was that the nurse administered 10 × 200mg tablets to the patient without any reference or confirmation that this was indeed what was intended. This use of amiodarone is at present outside the product licence and would not have been described in any of the literature available on the ward.

The patient subsequently died, but at the time of writing no causal effect from this high dose of amiodarone had been established.

Case Two

A female patient aged approximately 65 years was prescribed 2,500 units of dalteparin sodium subcutaneously once a day as part of the DVT prophylaxis protocol. The prescribed dose was misread and two nurses checking each other gave five pre-filled syringes, i.e. 25,000 units, to the patient in error. So much heparin was required that another patient's supply had to be used as well and the error came to light when the ward made a request to pharmacy for 25,000 unit doses of dalteparin. When the error was discovered the patient's coagulation status was checked immediately and she fortunately came to no harm.

Comment

It seems inconceivable that such high numbers of dose units could be administered to patients without the nurses involved at least querying that something might be wrong.

Taken from: *Pharmacy in Practice* 2001; **6**: 194.

PROBLEMS

Question 13 You need to give 1 g of erythromycin orally. You have erythromycin suspension 250 mg in 5 mL.

How much of the suspension do you need to give?

Question 14 You need to give a patient 62.5 micrograms of digoxin orally. You have digoxin liquid containing 50 micrograms/mL.

How much do you need to draw up?

Question 15 If Oramorph® concentrate 100 mg/5 ml is used to give a dose of 60 mg for breakthrough pain, what volume is required?

Question 16 You have pethidine injection 100 mg in 2 mL. The patient is prescribed 75 mg.

How much do you draw up?

Question 17 You need to give ranitidine liquid at a dose of 2 mg/kg to a 9-year-old child weighing 23 kg.

You have a 150 mg in 10 mL liquid. How much do you need to give for each dose?

Question 18 You need to give a dose of trimethoprim suspension to a child weighing 18.45 kg at a dose of 4 mg/kg. You have trimethoprim suspension 50 mg in 5 mL.

What dose do you need to give and how much of the suspension do you need?

Question 19 Ciclosporin (cyclosporin) has been prescribed to treat a patient with severe rheumatoid arthritis. The oral dose is 2.5 mg/kg daily in two divided doses. The patient weighs 68 kg. Ciclosporin (cyclosporine) is available in 10 mg, 25 mg, 50 mg and 100 mg capsules. What dose is required and which strength of capsules would you need to give?

Question 20 You need to give aciclovir (acyclovir) as an infusion at a dose of 5 mg/kg every 8 hours. The patient weighs 76 kg and aciclovir (acyclovir) is available as 250 mg vials. How many vials do you need for each dose?

Question 21 A 50 kg woman is prescribed aminophylline as an infusion at a dose of 0.5 mg/kg/hour.

Aminophylline injection comes as 250 mg in 10 mL ampoules.

How much is required if the infusion is to run for 12 hours?

Question 22 You need to prepare an infusion of co-trimoxazole at a dose of 120 mg/kg/day in four divided doses for a patient weighing 68 kg.

Co-trimoxazole is available as 5 mL ampoules at a strength of 96 mg/mL.

i) What volume of co-trimoxazole do you need for each dose?

ii) How many ampoules do you need for each dose?

DISPLACEMENT VALUES OR VOLUMES

Dry powder injections need to be reconstituted with a diluent before they are used. Sometimes the final volume of the injection will be greater than the volume of liquid that was added to the powder. This volume difference is called the injection's displacement value. What is displacement? If you take ordinary salt and dissolve it in some water, the resultant solution will have a greater volume than before. The salt appears to 'displace' some water increasing the volume.

Antibiotic suspensions are good examples to illustrate displacement. For example, to make up 100mL of amoxicillin (amoxycillin) suspension, only 68 mL of water needs to be added. The amoxicillin (amoxycillin) powder must therefore displace 32mL of water, i.e. 100 – 68 = 32mL.

Is displacement important in medicine? For most patients this does not matter because the whole vial is administered. However it can be very important when you want to give a dose that is less than the total contents of the vial – a frequent occurrence in paediatrics and neonatology.

The volume of the final solution must be considered when calculating the amount to withdraw from the vial. The total volume may be increased significantly and, if this is not taken into account, significant errors in dosage may occur, especially when small doses are involved as with neonates. For example:

Cefotaxime at a dose of 50 mg/kg 12-hourly for a baby weighing 3.6 kg.

Therefore dose required = 50 × 3.6 = 180mg

Displacement volume for cefotaxime = 0.2mL for a 500mg vial.

Therefore you need to add 1.8mL (2 – 0.2) water for injection to give a final concentration of:

500mg in 2mL

Thus the required dose in mL is:

$$180\ \text{mg} = \frac{2}{500} \times 180 = 0.72\text{mL}$$

If the displacement volume is not taken into account, then you will have:

500 mg in 2.2 mL (2 mL + 0.2 mL displacement volume)

You worked out earlier that the dose required is 180 mg and this is equal to 0.72 mL (assuming you have 500 mg in 2 mL). But now 0.72 mL contains:

$$\frac{500}{2.2} \times 0.72 = 164\text{mg}$$

(since the actual volume you have is 2.2mL and not 2mL). Thus if the displacement volume is not taken into account, then the amount drawn up is 164mg and not 180mg as required. Displacement values will depend on the medicine, the manufacturer and its strength. Information on a medicine's displacement value is usually stated in the relevant drug information sheets, in paediatric dosage books, or can be obtained from your Pharmacy Department. Calculating doses using displacement volumes:

volume to be added = diluent volume − displacement volume

For example, for benzylpenicillin:

Dose required = 450mg

Displacement volume = 0.4mL per 600mg vial

Total diluted volume = 5mL

Volume of diluent to be added = 5 – 0.4 = 4.6mL

Final concentration = 120mg/mL

Volume required to deliver 450mg dose = 3.75mL

PROBLEMS

Work out the following dosages, not forgetting to take into account displacement values if necessary.

Question 23 You need to give a 4-month-old child 350 mg ceftriaxone IV daily. You have a 1 g vial that needs to be reconstituted to 10 mL with Water for Injections. Displacement value = 0.8mL for 1 g.

i) Taking into account displacement volumes, what volume should you add to the vial?

ii) How much do you need to draw up for a 350mg dose?

Question 24 You need to give cefotaxime IV to a 5-year-old child weighing 18 kg at a dose of 150 mg/kg/day in four divided doses. You have a 1 g vial that needs to be reconstituted to 4 mL with Water for Injections. Displacement value = 0.5mL for 1 g.

i) Taking into account displacement volumes, what volume should you add to the vial?

ii) How much do you need to draw up for each dose?

MOLES AND MILLIMOLES

OBJECTIVES

At the end of this chapter, you should be familiar with the following:

- Moles and millimoles
- Millimoles and micromoles
- Calculations involving moles and millimoles

  Conversion of milligrams to millimoles

  Conversion of percentage strength (% w/v) to millimoles

- Molar solutions and molarity

KEY POINTS

Moles

- The **mole** is a unit used by chemists to count atoms and molecules.
- A mole of any substance is the amount that contains the same number of particles of the substance as there are atoms in 12 g of carbon (C^{12}) – known as Avogadro's number.
- For elements or atoms:

  one mole = the **atomic mass** in grams
- For molecules:

  one mole = the **molecular mass** in grams

Millimoles

- Moles are too big for everyday use, so the unit **millimoles** is used.
- One millimole is equal to one-thousandth of a mole.
- If one mole is the atomic mass or molecular mass in grams, it follows that:

  one **millimole** = the **atomic mass** or **molecular mass** in **milligrams**

Conversion of Milligrams (mg) to Millimoles (mmol)

mg/mL to millimoles

$$\text{total number of millimoles} = \frac{\text{strength in mg/mL}}{\text{mg of substance containing 1 mmol}} \times \text{volume (in mL)}$$

Conversion of percentage strength (% w/v) to millimoles

$$\text{total number of mmol} = \frac{\text{percentage strength (\% w/v)}}{\text{mg of substance containing 1 mmol}} \times 10 \times \text{volume (mL)}$$

Molar Solutions and Molarity

- Molarity is a term used in chemistry to describe concentrations:

When **one mole** of a substance is dissolved in **one litre** of solution, it is known as a **one molar** (1 M) solution.

- A one molar (1 M) solution has one mole of the substance dissolved in each litre of solution (equivalent to 1 mmol per mL).
- If 2 moles of a substance are made up to 1 litre (or 1 mole to 500 mL), the solution is said to be a 2 M solution.

INTRODUCTION

Daily references may be made to moles and millimoles in relation to electrolyte levels, blood glucose, serum creatinine or other blood results with regard to patients. These are measurements carried out by chemical pathology and the units used are usually millimoles or micromoles. The millimole unit is also encountered with infusions when electrolytes have been added. For example:

Mr J. Brown Sodium = 138 mmol/L

'The infusion contains 20mmol potassium chloride.' Before you can interpret such results or amounts, you will need to be

familiar with this rather confusing unit: the mole.

This section will explain what moles and millimoles are, and how to do calculations involving millimoles.

WHAT ARE MOLES AND MILLIMOLES?

It is important to know what moles and millimoles are and how they are derived. However, the concept of moles and millimoles is difficult to explain and to understand; you need to be familiar with basic chemistry. Chemists are concerned with atoms, ions and molecules. These are too small to be counted individually, so the mole is the unit used by chemists to make counting and measuring a lot easier.

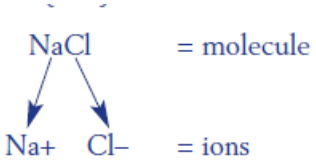
So what is a mole? Just as the word 'dozen' represents the number 12, the mole also represents a number – 6 × 1023. This number can represent atoms, ions or molecules, e.g. 1 mole of atoms is 6 × 1023 atoms.The mole is the SI unit for the amount of a substance and so one mole also represents the relative atomic, molecular or ionic mass in grams.

The atomic mass of potassium is 39; so 1 mole of potassium has a mass of 39 g (which is the same as saying that 6 × 1023 atoms of potassium have a total mass of 39 g).

Similarly, the molecular weight of sodium chloride is 58.5; therefore a mole of sodium chloride has a mass of 58.5 g and 2 moles of sodium chloride have a mass of 117 g (2 × 58.5).

Chemists are not the only the people who 'count by weighing'. Bank clerks use the same idea when they count coins by weighing them. For example, one hundred 1p coins weigh 356 g; it is quicker to weigh 356 g of 1p coins than to count a hundred coins.

Now consider a single molecule of sodium chloride (NaCl) which consists of one sodium ion (Na^+) and one chloride ion ($Cl^–$).



Since moles can refer to ions as well as molecules, it can be seen that one mole of sodium chloride contains one mole of sodium ions and one mole of chloride ions. From tables (see the end of this section), the relative ionic masses are:

sodium (Na) 23

chloride (Cl) 35.5

The molecular mass is the sum of the ionic masses:

sodium (Na) 23

chloride (Cl) 35.5

58.5 = molecular mass of NaCl

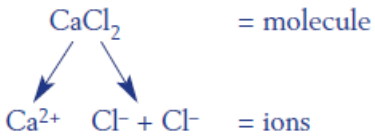
So we can say that:

1 mole of sodium ion weighs 23 g;

1 mole of chloride ion weighs 35.5 g;

1 mole of sodium chloride weighs 58.5 g

Next consider calcium chloride (CaCl2). Each molecule consists of **one** calcium ion and **two** chloride ions. The '2' after the 'Cl' means **two** ions of chlorine:



Themolecular mass of calcium chloride is 147.The reason why the molecular mass does not always equal the sum of the atomic masses of the individual ions is because water forms a part of each calcium chloride molecule.

The molecule is actually CaCl2·2H2O. From the molecular formula and knowledge of the atomic weights it can be seen that calcium chloride contains:

1 mole of Ca = 40g

2 moles of Cl = 71 g

2 moles of H2O, each mole of water = 18 g; 2 × 18 = 36 g

So adding everything together:

(40 + 71 +36) = 147 i.e. the mass of 1 mole CaCl2 is 147g.

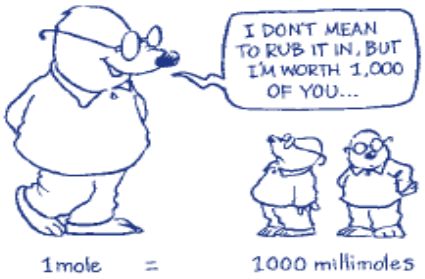
TABLE 7.1 ATOMIC AND MOLECULAR MASS	
ATOM/MOLECULE	**MASS**
Calcium (Ca)	40
Calcium chloride	147
Calcium gluconate	448.5
Carbon (C)	12
Chloride (Cl)	35.5
Dextrose/glucose	180
Hydrogen (H)	1
Magnesium (Mg)	24
Magnesium chloride	203
Magnesium sulphate	246.5
Oxygen (O)	16
Potassium (K)	39
Potassium chloride	74.5
Sodium (Na)	23
Sodium bicarbonate	84
Sodium chloride	58.5
Sodium citrate	294
Sodium phosphate	358
Water (H_2O)	18

MILLIMOLES AND MICROMOLES

Moles are, however, too big for everyday use in medicine, so millimoles and micromoles are used.

One millimole is equal to one-thousandth of a mole

One micromole is equal to one-thousandth of a millimole



It follows that:

1 mole contains 1,000 millimoles (mmol)

1 millimole contains 1,000 micromoles (mcmol)

So, in the above explanation, you can substitute millimoles for moles and

milligrams for grams. For our purposes:

Sodium chloride	would give →	sodium	+	chloride
1 mole or		**1 mole** or		**1 mole** or
1 millimole		**1 millimole**		**1 millimole**
58.5 g or		23 g or		
	35.5 g or			
58.5 mg		23 mg		
	35.5 mg			

The following are examples and problems to see if you understand the concept of millimoles. It is unlikely that you will encounter these types of calculations on the ward, but it is useful to know how they are done and they can be used for reference if necessary.



CALCULATIONS INVOLVING MOLES AND MILLIMOLES

Conversion of milligrams (mg) to millimoles (mmol)

Sometimes it may be necessary to calculate the number of millimoles in an infusion or injection or to convert mg/litre to mmol/litre.

WORKED EXAMPLE

How many millimoles of sodium are there in a 500 mL infusion containing 1.8 mg/mL sodium chloride?

STEP ONE

As already stated: 1 millimole of sodium chloride yields 1 millimole of sodium. So it follows that the amount (in milligrams) equal to 1 millimole of sodium chloride will give 1 millimole of sodium.

In this case, calculate the total amount (in milligrams) of sodium chloride and convert this to millimoles to find out the number of millimoles of sodium.

STEP TWO

Calculate the total amount of sodium chloride.

You have an infusion containing 1.8 mg/mL.

Therefore in 500 mL, you have:

$$1.8 \times 500 = 900 \text{ mg sodium chloride}$$

STEP THREE

From tables: molecular mass of sodium chloride (NaCl) = 58.5.

So 1 millimole of sodium chloride (NaCl) will weigh 58.5 mg and this amount will give 1 millimole of sodium (Na).

STEP FOUR

Next calculate the number of millimoles in the infusion. First work out the number of millimoles for 1 mg of sodium chloride, and then the number for the total amount.

$$58.5\text{ mg sodium chloride will give 1 millimole of sodium.}$$

$$1\text{ mg will give } \frac{1}{58.5}\text{ millimoles of sodium.}$$

$$\text{So, 900 mg will give } \frac{1}{58.5} \times 900 = 15.38\text{ mmol (or 15 mmol approx.)}$$

ANSWER: There are 15.4 mmol (approximately 15 mmol) of sodium in a 500 mL infusion containing 1.8 mg/mL sodium chloride.

Alternatively, a formula can be used:

$$\text{total number of millimoles} = \frac{\text{mg/mL}}{\text{mg of substance containing 1 mmol}} \times \text{volume(mL)}$$

where, in this case

$$\text{mg/mL} = 1.8$$

$$\text{mg of substance containing 1 mmol} = 58.5$$

$$\text{volume (mL)} = 500$$

Substituting the numbers in the formula:

$$\frac{1.8}{58.5}\text{ mmol} \times 500 = 15.38\text{ mmol (or 15 mmol approx.)}$$

ANSWER: There are 15.4 mmol (approximately 15 mmol) of sodium in a 500 mL infusion containing 1.8 mg/mL sodium chloride.

PROBLEMS

Question 1 How many millimoles of sodium are there in a 500 mL infusion containing 27 mg/mL sodium chloride? (MM sodium chloride = 58.5)

Question 2 How many millimoles of sodium are there in a 10 mL ampoule containing 300 mg/mL sodium chloride? (MM sodium chloride = 58.5)

Question 3 How many millimoles of sodium, potassium and chloride are there in a 500 mL infusion containing 9 mg/mL sodium chloride and potassium chloride 3 mg/mL? (MM sodium chloride = 58.5; MW potassium chloride = 74.5)

Conversion of percentage strength (% w/v) to millimoles

Sometimes it may be necessary to convert percentage strength to the number of millimoles. This is very similar to the previous example. All you need to do is to convert the percentage strength to the number of milligrams in the required volume, then follow the steps as before.

WORKED EXAMPLE

How many millimoles of sodium are in 1 litre of sodium chloride 0.9% infusion?

STEP ONE

We know that 1 millimole of sodium chloride will give 1 millimole of sodium.

So, the amount (in milligrams) equal to 1 millimole of sodium chloride will give 1 millimole of sodium.

STEP TWO

To calculate the number of milligrams in 1 millimole of sodium chloride, either refer to tables or work from first principles using atomic masses.

From tables: molecular mass of sodium chloride (NaCl) = 58.5.

So 1 millimole of sodium chloride (NaCl) will weigh 58.5 mg and this amount will give 1 millimole of sodium (Na).

STEP THREE

Calculate the total amount of sodium chloride present:

0.9% = 0.9 g in 100mL

= 900mg in 100mL

Thus in a 1 litre (1,000mL) infusion, the amount is:

$$\frac{900}{100} \times 1{,}000 = 9{,}000\,\text{mg or } 9\,\text{g}$$

STEP FOUR

From Step Two, it was found that:

58.5mg sodium chloride will give 1 millimole of sodium.

So it follows that:

$$1\,\text{mg of sodium chloride will give } \frac{1}{58.5} \text{ millimoles of sodiu}$$

$$\text{So, } 9{,}000\,\text{mg sodium chloride will give}$$

$$\frac{1}{58.5} \times 9{,}000 = 153.8\ (154)\,\text{mmol of sodium}$$

ANSWER: 1 litre of sodium chloride 0.9% infusion contains 154 mmol of sodium (approx.).

A formula can be devised:

$$\text{mmol} = \frac{\text{percentage strength (\% w/v)}}{\text{mg of substance containing 1 mmol}} \times 10 \times \text{volume (mL)}$$

In this example:

$$\text{percentage strength (\% w/v)} = 0.9$$
$$\text{mg of substance containing 1 mmol} = 58.5$$
$$\text{volume (mL)} = 1{,}000$$

Then 10 simply converts percentage strength (g/100 mL) to mg/mL (everything in the same units).
Substituting the numbers in the formula:

$$\frac{0.9}{58.5} \times 10 \times 1{,}000 = 153.8\ (154)\ \text{mmol of sodium}$$

ANSWER: 1 litre of sodium chloride 0.9% infusion contains 154 mmol of sodium (approx.).

PROBLEMS

Question 5 How many millimoles of sodium are there in a 1 litre infusion of glucose 4% and sodium chloride 0.18%? (MM sodium chloride = 58.5; glucose = 180.)

Question 6 How many millimoles of calcium and chloride are there in a 10mL ampoule of calcium chloride 10%? (MM calcium chloride = 147)

Note: calcium chloride = CaCl2

Question 7 How many millimoles of calcium are there in a 10 mL ampoule of calcium gluconate 10%? (MM calcium gluconate = 448.5)

Question 8 How many millimoles of sodium are there in a 200mL infusion of sodium bicarbonate 8.4%? (MM sodium bicarbonate = 84)

INFUSION RATE CALCULATIONS

OBJECTIVES

At the end of this chapter you should be familiar with the following:

• Drip rate calculations (drops/min)

• Conversion of dosages to mL/hour

• Conversion of mL/hour back to a dose

• Calculating the length of time for IV infusions

KEY POINTS

Drip Rate Calculations (drops/min)

- In all drip rate calculations, you have to remember that you are simply converting a volume to drops (or vice versa) and hours to minutes.

$$\text{drops/min} = \frac{\text{drops/mL of the giving set} \times \text{volume of the infusion (mL)}}{\text{number of hours the infusion is to run} \times 60}$$

Giving sets

- The **standard giving set (SGS)** has a drip rate of **20 drops per mL** for clear fluids (i.e. sodium chloride, glucose) and **15 drops per mL** for blood.
- The **micro-drop giving set** or burette has a drip rate of **60 drops per mL**.

Conversion of Dosages to mL/hour

- In this type of calculation, it is best to convert the dose required to a volume in millilitres.
- Doses expressed as mcg/kg/min:

$$\text{mL/hour} = \frac{\text{volume to be infused} \times \text{dose} \times \text{weight} \times 60}{\text{amount if drug} \times 1{,}000}$$

60 converts minutes to hours.
1,000 converts mcg to mg.

- If doses are expressed in milligrams, then there is no need to divide by 1,000.
- If doses are expressed as a total dose, i.e. dose/min, there is no need to multiply by the patient's weight.

Conversion of mL/hour Back to a Dose

- Sometimes it may be necessary to convert *mL/hour* back to the dose in *mg/min* or *mcg/min* and *mg/kg/min* or *mcg/kg/min*.

$$\text{mcg/kg/min} = \frac{\text{rate (mL/hour)} \times \text{amount of drug} \times 1{,}000}{\text{weight (kg)} \times \text{volume (mL)} \times 60}$$

1,000 converts mcg to mg.
60 converts minutes to hours.

- If doses are expressed in terms of milligrams, then there is no need to multiply by 1,000.
- If doses are expressed as a total dose, i.e. dose/min, there is no need to divide by the patient's weight.

Calculating the Length of Time for IV Infusions

- Sometimes it may be necessary to calculate the number of hours an infusion should run at a specified rate. Also, it is a good way of checking your calculated drip rate for an infusion.

Manually controlled infusions

$$\text{number of hours the infusion is to run} = \frac{\text{volume of the infusion}}{\text{rate (drops/min)} \times 60} \times \text{drip rate of giving set}$$

Infusion or syringe pumps

$$\text{number of hours the infusion is to run} = \frac{\text{volume of the infusion}}{\text{rate (mL/hour)}}$$

INTRODUCTION

With infusions, there are two types of infusion rate calculations to be considered: those involving **drops/min** and those involving **mL/hour**.

The first (drops/min) is mainly encountered when infusions are given under gravity as with fluid replacement. The second (mL/hour) is encountered when infusions have to be given accurately or in small volumes using infusion or syringe pumps – particularly if drugs have to be given as infusions.

DRIP RATE CALCULATIONS (drops/min)

To set up a manually controlled infusion accurately, you need to be able to count the number of drops per minute. To do this, you have to calculate the volume to be infused in terms of drops.This in turn depends upon the giving or administration set being used.

Giving sets

There are two giving sets:

• The **standard giving set (SGS)** has a drip rate of **20 drops per mL** for clear fluids (i.e. sodium chloride, glucose) and **15 drops per mL** for blood.

• The **micro-drop giving set** or burette has a drip rate of **60 drops per mL**. The drip rate of the giving set is always written on the wrapper if you are not sure.

In all drip rate calculations, you have to remember that you are simply converting a volume to drops (or vice versa) and hours to minutes.

WORKED EXAMPLE

1 litre of sodium chloride 0.9% ('normal saline') is to be given over 8 hours: what drip rate is required using a standard giving set (SGS), 20 drops/mL?

STEP ONE

First convert the volume to a number of drops. To do this, multiply the volume of the infusion by the number of 'drops per mL' for the giving set, i.e.

1 litre = 1,000 mL, so it will be

$$1{,}000 \times 20 = 20{,}000 \text{ drops}$$

You have just calculated the number of drops to be infused.

STEP TWO

Next convert hours to minutes by multiplying the number of hours over which the infusion is to be given by 60 (60 minutes = 1 hour).

$$8 \text{ hours} = 8 \times 60 = 480 \text{ minutes}$$

Now everything has been converted in terms of drops and minutes, i.e. what you want for your final answer.

If the infusion is being given over a period of minutes, then obviously there is no need to convert from hours to minutes.

STEP THREE

Write down what you have just calculated, i.e. the total number of drops to be given over how many minutes.

20,000 drops to be given over 480 minutes

STEP FOUR

Calculate the number of drops per minute by dividing the number of drops by the number of minutes:

$$\frac{20{,}000}{480} = 41.67 \text{ drops/min}$$

Since it is impossible to have part of a drop, round up or down to the nearest whole number:

$$41.67 \approx 42 \text{ drops/min}$$

ANSWER: To give a litre (1,000 mL) of sodium chloride 0.9% ('normal saline') over 8 hours using a standard giving set (20 drops/mL), the rate will have to be 42 drops/min.

A formula can be used:

$$\text{drops/min} = \frac{\text{drops/mL of the giving set} \times \text{volume of the infusion (mL)}}{\text{number of hours the infusion is to run} \times 60}$$

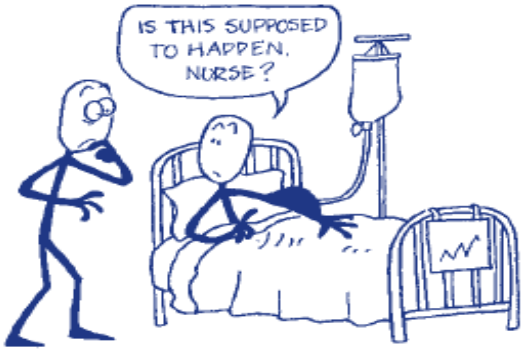
where in this case:

drops/mL of the giving set = 20 drops/mL (SGS)
volume of the infusion (in mL) = 1,000 mL
number of hours the infusion is to run = 8 hours
60 = number of minutes in an hour (converts hours to minutes)

Substituting the numbers into the formula:

$$\frac{20 \times 1{,}000}{8 \times 60} = 41.67 \text{ drops/min (42 drops/min, approx.)}$$

ANSWER: To give a litre (1,000 mL) of sodium chloride 0.9% ('normal saline') over 8 hours using a standard giving set (20 drops/mL), the rate will have to be 42 drops/min .



PROBLEMS

Work out the drip rates for the following infusions:

Question 1 500mL of sodium chloride 0.9% over 6 hours

Question 2 1 litre of glucose 4% and sodium chloride 0.18% over 12 hours

Question 3 1 unit of blood (500 mL) over 6 hours

CONVERSION OF DOSAGES TO mL/hour

Dosages can be expressed in various ways: *mg/min* or *mcg/min* and *mg/kg/min* **or** *mcg/kg/min*; and it may be necessary to convert to *mL/hour* (when using infusion pumps).

The following example shows the various steps in this type of calculation, and this can be adapted for any dosage to infusion rate calculation.

WORKED EXAMPLE

You have an infusion of dopamine 800 mg in 500 mL. The dose required is 2 mcg/kg/min for a patient weighing 68 kg. What is the rate in mL/hour?

STEP ONE

When doing this type of calculation, you need to convert the dose required to a volume in mL and minutes to hours.

STEP TWO

First calculate the dose required:

$$\text{dose required} = \text{patient's weight} \times \text{dose prescribed}$$
$$= 68 \times 2 = 136 \text{ mcg/min}$$

If the dose is given as a total dose and not on a weight basis, then miss out this step.

STEP THREE

The dose is 136 mcg/min. The final answer needs to be in terms of hours, so multiply by 60 to convert minutes into hours:

$$136 \times 60 = 8{,}160 \text{ mcg/hour}$$

Convert mcg to mg by dividing by 1,000:

$$= 8.16 \text{ mg/hour}$$

STEP FOUR

The next step is to calculate the volume for the dose required.

Calculate the volume for 1 mg of drug:

You have: 800 mg in 500 mL:

$$1 \text{ mg} = \qquad = 0.625 \text{ mL}$$

STEP FIVE

Thus for the dose of 8.16 mg, the volume is equal to:

$$8.16 \text{ mg} = 0.625 \times 8.16 = 5.1 \text{ mL/hour}$$

ANSWER: The rate required is 5.1 mL/hour.

A formula can be derived:

mL/hour =

In this case:

total volume to be infused = 500 mL
total amount of drug (mg) = 800 mg
dose = 2 mcg/kg/min
patient's weight = 68 kg
60 converts minutes to hours
1,000 converts mcg to mg

Substituting the numbers into the formula:

= 5.1 mL/hour

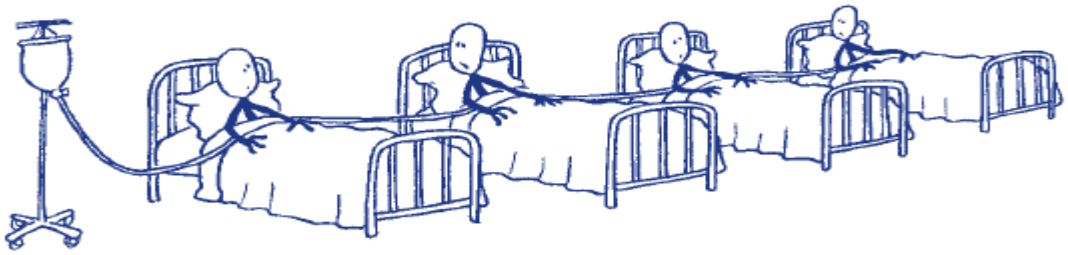
ANSWER: The rate required is 5.1 mL/hour.

If the dose is given as a total dose and not on a weight basis, then the patient's weight is not needed.

mL/hour =

Note:

- If doses are expressed in terms of milligrams, then there is no need to divide by 1,000.
- If doses are expressed as a total dose, i.e. dose/min, there is no need to multiply by the patient's weight.



I KNOW THEY'RE SUPPOSED TO CUT COSTS, BUT THIS IS RIDICULOUS!

TABLE OF INFUSION RATES (ML/HOUR)

Time / Vol	MINUTES					HOURS										
	10	15	20	30	40	1	2	3	4	6	8	10	12	16	18	24
10 mL	60	40	30	20												
20 mL	120	80	60	40												
30 mL				60												
40 mL				80	60	40										
50 mL				100	75	50										
60 mL				120	90	60										
80 mL				160	120	80										
100 mL				200	150	100	50	33	25	17						
125 mL				250	188	125	63	42	31	21						
150 mL				300	225	150	75	50	38	25						
200 mL				400	300	200	100	67	50	33	25	20	17	13	11	8
250 mL							125	83	63	42	31	25	21	16	14	10
500 mL									125	83	63	50	42	31	28	21
1000 mL									250	167	125	100	83	63	56	42

Rates given in the table have been rounded up or down to give whole numbers.

How to use the table

If you need to give a 250 mL infusion over 8 hours, then to find the infusion rate (mL/hour) go down the left-hand (Vol) column until you reach 250 mL; then go along the top (Time) line until you reach 8 (for 8 hours). Then read off the corresponding infusion rate (mL/hour). In this case it is 31mL/hour.

PROBLEMS

Question 4 You have a 500 mL infusion containing 50 mg nitroglycerin. A dose of 10 mcg/min is required. What is the rate in mL/hour?

Question 5 You are asked to give 500 mL of lidocaine (lignocaine) 0.2% in glucose at a rate of 2 mg/min. What is the rate in mL/hour?

Question 6 You have an infusion of dopamine 800mg in 500 mL. The dose required is 3 mcg/kg/min for a patient weighing 80 kg. What is the rate in mL/hour?

Question 7 A patient with chronic obstructive pulmonary disease (COPD) is to have a continuous infusion of aminophylline. The patient weighs 63 kg and the dose to be given is 0.5 mg/kg/hour over 12 hours. Aminophylline injection comes as 250mg in 10mL ampoules and should be given in a 500mL infusion bag.

(i) What dose and volume of aminophylline are required?

(ii) What is the rate in mL/hour?

Question 8 You need to give aciclovir (acyclovir) as an infusion at a dose of 5 mg/kg every 8 hours. The patient weighs 86 kg and acyclovir (acyclovir) is available as 500 mg vials. Each vial needs to be reconstituted with 20mL Water for Injection and diluted further to 100 mL. The infusion should be given over 60 minutes.

(i) What dose and volume of aciclovir (acyclovir) are required for one dose?

(ii) What is the rate in mL/hour for each dose?

Question 9 Glyceryl trinitrate is to be given at a rate of 150 mcg/min. You have an infusion of 50mg in 50mL glucose 5%. What is the rate in mL/hour?

Question 10 You have an infusion of dobutamine 250mg in 50 mL. The dose required is 6 mcg/kg/min and the patient weighs 75 kg. What is the rate in mL/hour?

Question 11 A patient with MRSA is prescribed IV vancomycin 1 g every 24 hours. After reconstitution a 500 mg vial of vancomycin should be diluted with infusion fluid to 5 mg/mL

CONVERSION OF mL/hour BACK TO A DOSE

Sometimes it may be necessary to convert *mL/hour* back to the dose: *mg/min* or *mcg/min* and *mg/kg/min* or *mcg/kg/min.* For example, you may need to check that an infusion pump is giving the correct dose. Nurses changing shifts, especially on the critical care wards, must check that the pumps are set correctly at the beginning of each shift.

WORKED EXAMPLE

An infusion pump containing 250 mg dobutamine in 50 mL is running at a rate of 3.5 mL/hour. Convert this to mcg/kg/min to check that the pump is set correctly. (Patient's weight = 73 kg)

STEP ONE

In this type of calculation, convert the volume being given to the amount of drug, and then work out the amount of drug being given per minute or per kilogram of the patient's weight.

STEP TWO

You have 250 mg of dobutamine in 50 mL.

First it is necessary to work out the amount in 1 mL:

250 mg in 50 mL

1 mL = mg = 5 mg (using the 'ONE unit' rule)

STEP THREE

The rate at which the pump is running is 3.5 mL/hour. You have just worked out the amount in 1 mL (Step Two)

3.5 mL/hour = 5 × 3.5 = 17.5 mg/hour

Now convert the rate (mL/hour) to the amount of drug being given over an hour.

STEP FOUR

The question asks for the dose in *mcg/kg/m* so convert milligrams to micrograms by multiplying by 1,000:

17.5 × 1,000 = 17,500 mcg/hour

STEP FIVE

Now calculate the rate per minute by dividing by 60 (converts hours to minutes):

= 291.67 mcg/min

STEP SIX

The final step in the calculation is to work out the rate according to the patient's weight (73 kg). (If the dose is not given in terms of the patient's weight, then miss out this final step.)

= 3.99 mcg/kg/min

This can be rounded up to 4 mcg/kg/min.

Now check your answer against the dose written on the drug chart to see if the pump is delivering the correct dose. If your answer does not match the dose written on the drug chart, then re-check your calculation. If the answer is still the same, then inform the doctor and, if necessary, calculate the correct rate.

A formula can be derived:

mcg/kg/min =

where in this case:

rate = 3.5 mL/hour
amount of drug (mg) = 250 mg
weight (kg) = 73 kg
volume (mL) = 50 mL
60 converts minutes to hours
1,000 converts mg to mcg

Substitute the numbers in the formula:

= 3.99 mcg/kg/min

This can be rounded up to 4 mcg/kg/min.

If the dose is given as a total dose and not on a weight basis, then the patient's weight is not needed:

mcg/min =

Note: If the dose is in terms of milligrams, then there is no need to multiply by 1,000 (i.e. delete from the formula).

PROBLEMS

Convert each of the following infusion pump rates to a mcg/kg/min dose:

Question 14 You have dopamine 200 mg in 50 mL and the rate at which the pump is running is 4 mL/hour. The prescribed dose is 3 mcg/kg/min. What dose is the pump delivering? (Patient′s weight = 89 kg) If the dose is wrong, at which rate should the pump be set?

Question 15 You have dobutamine 250 mg in 50 mL and the rate at which the pump is running is 5.4 mL/hour. The prescribed dose is 6 mcg/kg/min. What dose is the pump delivering? (Patient′s weight = 64 kg) If the dose is wrong, at which rate should the pump be set?

Question 16 You have dopexamine 50 mg in 50 mL and the rate at which the pump is running is 28 mL/hour. The prescribed dose is 6 mcg/kg/min. What dose is the pump delivering? (Patient′s weight = 78 kg) If the dose is wrong, at which rate should the pump be set?

CALCULATING THE LENGTH OF TIME FOR IV INFUSIONS

It may sometimes be necessary to calculate the number of hours an infusion should run at a specified rate. Also, it is a good way of checking your calculated drip rate or pump rate for an infusion. For example: you are asked to give a litre of 5% glucose over 8 hours. You have calculated that the drip rate should be 42 drops/min (using a standard giving set: 20 drops/mL) or 125 mL/hour for a pump. To check your answer, you can calculate how long the infusion should take at the calculated rate. If your answers do not correspond (the answer

should be 8 hours), then you have made an error and should re-check your calculation. Alternatively, you can use this type of calculation to check the rate of an infusion already running. For example: if an infusion is supposed to run over 6 hours, and the infusion is nearly finished after 4 hours, you can check the rate by calculating how long the infusion should take using that drip rate or the rate set on the pump. If the calculated answer is less than 6 hours, then the original rate was wrong and the doctor should be informed.

WORKED EXAMPLE

MANUALLY CONTROLLED INFUSIONS

The doctor prescribes 1 litre of 5% glucose to be given over 8 hours.

The drip rate for the infusion is calculated to be 42 drops/min. You wish to check the drip rate; how many hours is the infusion going to run? (SGS = 20 drops/mL)

In this calculation, you first convert the volume being infused to drops; then calculate how long it will take at the specified rate.

STEP ONE

First, convert the volume to drops by multiplying the volume of the infusion (in mL) by the number of drops/mL for the giving set:

volume of infusion = 1 litre = 1,000 mL

1,000 × 20 = 20,000 drops

STEP TWO

From the rate, calculate how many minutes it will take for 1 drop:

42 drops per minute

1 drop = min

STEP THREE

Calculate how many minutes it will take to infuse the total number of drops:

1 drop = min

20,000 drops = × 20,000 = 476 min

STEP FOUR

Convert minutes to hours by dividing by 60.

476 min = = 7.93 hours

How much is 0.93 of an hour?
Multiply by 60 to convert part of an hour back to minutes:

0.93 × 60 = 55.8 = 56 min (approx.)

ANSWER: 1 litre of glucose 5% at a rate of 42 drops/min will take approximately 8 hours to run (7 hours and 56 minutes).

A formula can be used:

number of hours the infusion is to run =

× drip rate of giving set

where in this case:

volume of the infusion = 1,000 mL
rate (drops/min) = 42 drops/min
drip rate of giving set = 20 drops/mL
60 converts minutes to hours

Substituting the numbers into the formula:

× 20 = 7.94 hours

Convert 0.94 hours to minutes: = 56 mins (approx.)

ANSWER: 1 litre of glucose 5% at a rate of 42 drops/min will take approximately 8 hours to run (7 hours 56 minutes).

WORKED EXAMPLE

INFUSION OR SYRINGE PUMPS

The doctor prescribes 1 litre of 5% glucose to be given over 8 hours. The rate for the infusion is calculated to be 125 mL/hour. You wish to check the rate; how many hours is the infusion going to run?

This is a simple calculation. You divide the total volume (in mL) by the rate to give the time over which the infusion is to run:

calculated rate = 125 mL/hour
volume = 1,000 mL (1 litre = 1000 mL)

= 8 hours

ANSWER: 1 litre of glucose 5% at a rate of 125 mL/hour will take 8 hours to run.

A simple formula can be used:

number of hours the infusion is to run =

where in this case:

volume of the infusion = 1,000 mL
rate (mL/hour) = 125 mL/hour

Substituting the numbers into the formula:

= 8 hours

ANSWER: 1 litre of glucose 5% at a rate of 125 mL/hour will take 8 hours to run.

PROBLEMS

Question 17 A 500 mL infusion of sodium chloride 0.9% is being given over 4 hours. The rate at which the infusion is being run is 42 drops/min. How long will the infusion run at the specified rate (SGS)?

Question 18 A 1 litre infusion of sodium chloride 0.9% is being given over 12 hours. The rate at which the infusion is being run is 83mL/hour. How long will the infusion run at the specified rate?

Question 19 Isosorbide dinitrate is to be given at a dose of 2 mg/hour. At what rate should the pump be set (mL/hour) to give this dose using a 0.05%w/v solution?

Question 20 A patient is receiving flucloxacillin 1g in 100mL sodium chloride 0.9% four times a day. Each 500mg vial of flucloxacillin contains1.13mmol sodium. How many mmol of sodium is the patient receiving in 24 hours? (Sodium chloride 0.9% contains 154 mmol sodium per litre.)

INFUSION DEVICES

OBJECTIVES

At the end of this chapter, you should be familiar with the following:

• Gravity devices

• Pumped systems

• Volumetric pumps

• Syringe pumps

• Patient-controlled analgesia (PCA)

• Anaesthesia pumps

• Pumps for ambulatory use

• Infusion device classification

KEY POINTS

Various devices are available.

Gravity Devices

- These depend entirely on gravity to drive the infusion; flow is measured by counting the drops.
- A gravity device should be considered only for low-risk infusions such as sodium chloride, dextrose saline and dextrose infusions.
- A gravity device should not be used for infusions:
  - containing potassium;
  - containing drug therapies requiring accurate monitoring or delivery of accurate volumes;
  - delivered to volume-sensitive patients.

Pumped Systems

These include the following different types:

Volumetric pumps

- Preferred pumps for medium and large flow rate and volume infusions; although some are designed specially to operate at low flow rates for neonatal use (not recommended for <5 mL/hour).

Syringe pumps

- These are used to administer drugs or infusions in small or medium volumes, and are calibrated at rates of 0.1 to 99 mL/hour (recommended for <5 mL/hour).
- Syringe pumps are used extensively where small volumes of highly concentrated drugs are required at low flow rates: usually in intensive care settings.

Patient-controlled analgesia (PCA) pumps

- These pumps are specifically designed for this purpose and are programmable to tailor analgesia for individual needs.

Anaesthesia pumps

- These are syringe pumps designed for use in anaesthesia or sedation and must be used *only* for this purpose; they are unsuitable for any other use. They should be restricted to operating and high dependency areas.

Pumps for ambulatory use

- Ambulatory pumps can be carried around by patients both in hospital and at home.

SOURCES AND INTERPRETATION OF DRUG INFORMATION

OBJECTIVES

At the end of this chapter, you should be familiar with the following:

• Sources of drug information

• Summary of product characteristics (SPC)

KEY POINTS

Sources of Drug Information

General

- *British National Formulary* (BNF) – website: http://www.bnf.org/bnf/
- *British National Formulary for Children* (BNFC) – website: http://bnfc.org/bnfc/
- *Electronic Medicines Compendium* – information provided by drug companies of their products – website: http://emc.medicines.org.uk/

Specialist books

- *Neonatal Formulary: Drug Use in Pregnancy & the First Year of Life* – current edition
- *Paediatric Formulary* – Guy's, St Thomas' and Lewisham Hospitals
- *Injectable Medicines Administration Guide* – Pharmacy Department, UCL Hospitals – Blackwell Publishing

Summary of Product Characteristics (SPC)

Information can be supplied by drug manufacturers in the form of a drug data sheet or summary of product characteristics (SPC). It provides essential information to ensure that the drug or medicine is used correctly, effectively and safely.

The information given in the SPC is also summarized in the package insert found inside the box. The package insert contains the basic information necessary for the administration and monitoring of the drug.

INTRODUCTION

Before administering any drug, you should have some idea of how that drug acts, its effect on the body and possible adverse effects. This is particularly true when dealing with parenteral products as they will have a more immediate and dramatic effect than oral preparations.

Basic drug information can be found in reference books or, increasingly, from web-based resources; many of the text-based resources are now also available on-line.

The following is a list of some sources of drug information. It is not meant to be exhaustive, but a list of resources that should be readily available on the ward, either as a book or via the Internet. The paediatric books should be available on paediatric wards; if not, they will be available in the hospital pharmacy.

SOURCES OF DRUG INFORMATION

General

- *British National Formulary* (BNF) – current edition
  (Latest edition = No. 57, March 2009)
  Website: http://www.bnf.org/bnf/
- *British National Formulary for Children* (BNFC) – current edition
  (Latest edition = 2008)
  Website: http://bnfc.org/bnfc/
- *Electronic Medicines Compendium* – information provided by drug companies of their products.
  Website: http://emc.medicines.org.uk/
- Drug package insert – if the drug is available in its original container, then there should be a package insert containing general information for that drug.

Specialist books

- *Neonatal Formulary: Drug Use in Pregnancy & the First Year of Life* – current edition (Latest edition = 5th, 2006)
  Northern Neonatal Network – Blackwell Publishing
  Updates are available from: http://www.blackwellpublishing.com/medicine/bmj/nnf5/
- *Paediatric Formulary* – current edition (Latest edition = 7th, 2005)
  Guy's, St Thomas' and Lewisham Hospitals

Your local hospital pharmacy department will be able to give advice on any aspects of dosing and administration of drugs.

Injectable Medicines Administration Guide – 2nd edition (2007). Pharmacy Departmnt, UCL Hospitals – Blackwell Publishing.